



Chat-First Data Analytics

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Abstract

Using natural language to drive data analysis is an old idea, but large language models (LLMs) have made it suddenly ubiquitous. A widely observed and concerning workflow has users upload a dataset to ChatGPT and ask for an analysis; the model generates and executes code on a remote virtual machine, producing results that are difficult to verify, difficult to reproduce, and incompatible with data-privacy requirements. This paper describes an alternative approach called *chat-first data analytics*, built on three principles: separate interpretation from computation, pre-validate methods rather than generate code, and transmit only metadata—not data. The language model interprets intent and selects tools; pre-implemented, validated algorithms produce results. **prompt2analytics** implements this approach as an econometrics and data analytics library in pure Rust, exposed through the Model Context Protocol (MCP) for integration with language models. The library covers regression, panel data, instrumental variables, difference-in-differences, discrete choice, time series, and over 200 additional methods—all validated against established R packages. Benchmarks across 384 configurations show a median speedup of $11\times$ over corresponding R implementations, with Rust faster in 89% of methods. End-to-end evaluation across six models shows that frontier models achieve 81–91% adequate rates on 96 prompts spanning tool selection, parameter extraction, and numerical correctness; a 3B-parameter model scores 80%, making fully local deployment practical. Local deployment via Ollama enables complete data isolation on consumer hardware.

Keywords: chat-first analytics, natural language interface, econometrics, Rust, MCP, local LLM, data privacy.

1. Introduction

The idea of talking to a computer to analyze data is not new. Natural language interfaces to databases date to the 1970s (Woods 1973; Codd 1974), and every generation of statistical software has tried to lower the barrier between analytical intent and software commands.

What *is* new is that the idea suddenly works well enough to be everywhere. Since late 2022, millions of users have been uploading datasets to ChatGPT and typing requests like “compare last quarter’s sales by region” (Hu 2023). The conversational interface to data analysis has gone from research curiosity to mass-market product in under two years.

The results are concerning. The model generates a Python script and executes it on a remote virtual machine—the user’s data now resides on a third-party server they do not control. Even when users copy generated code back to their own machine, the problems persist: generated code may compute the wrong quantity without raising an error (Ludwig, Mullainathan, and Rambachan 2025), and the same prompt yields different code as models update—making results irreproducible. Section 2 examines these failure modes in detail.

The core idea—natural language as interface—is sound. The architecture is not. These problems are not inherent to conversational data analysis; they arise from a specific design choice: delegating computation to generated code. This paper describes an alternative approach we call *chat-first data analytics*, built on three principles:

1. **Interpretation and computation are separated.** The language model selects methods and extracts parameters from natural language; pre-implemented, validated algorithms perform the actual computation.
2. **Methods are pre-validated, not generated.** Correctness is established once, through systematic validation against reference implementations, rather than hoped for anew with each request.
3. **Data stays local; only structure travels.** The language model receives column names, data types, and summary statistics—not the data itself. For complete isolation, the language model can run locally too.

These principles address each concern directly. Correctness follows from using validated implementations rather than generated code. Privacy is protected by keeping data on the user’s machine. Reproducibility is achieved because the same tool invocation produces identical results—the stochastic element of the language model affects only interpretation, not computation.

The target users are researchers and analysts who know what analysis they want but prefer not to write the code themselves: graduate students running standard econometric models for the first time, applied researchers iterating on specifications during exploratory work, and data analysts in organizations where privacy regulations prohibit cloud processing. For expert users fluent in R or Stata, the system offers faster iteration on routine tasks and a local privacy guarantee; for less experienced users, it lowers the barrier to correctly specified analyses—though statistical literacy remains necessary to interpret results (Section 2).

This paper presents **prompt2analytics** (**p2a**), an implementation of chat-first data analytics. The software includes an econometrics and data analytics library in pure Rust, exposed through the Model Context Protocol (MCP) for integration with language models. The implementation covers ordinary least squares (OLS) with robust standard errors, panel data models with fixed effects (FE) and random effects (RE), instrumental variables (IV), difference-in-differences (DiD), discrete choice, time series, and over 200 additional methods—all validated against established R packages.

prompt2analytics is available as open-source software under the MIT license at <https://github.com/umatter/prompt2analytics>. Supplementary materials—including all 96 evaluation prompts, benchmark scripts for both R and Rust, and validation test suites—are located in the repository’s `paper/` directory. Appendix D provides reproduction instructions.

The remainder of the paper is organized as follows. Section 2 motivates chat-first analytics by examining the failure modes of current approaches and situating the three principles within existing work. Section 3 describes the software implementation. Section 4 demonstrates the system through worked examples including realistic failure cases. Section 5 evaluates tool selection accuracy and numerical correctness. Section 6 presents performance benchmarks and local deployment configurations, and Section 7 discusses limitations and future directions.

2. Background and design principles

The introduction identified three concerns with using language models for data analysis: correctness, privacy, and reproducibility. This section examines these failure modes more carefully, then states the three principles that define chat-first analytics.

2.1. How LLM-assisted analysis goes wrong

The appeal of typing “regress wages on education with robust standard errors” into ChatGPT is obvious. The problems are less visible. They fall into four categories.

First, LLM-generated code often runs without error but computes the wrong quantity. Requesting “clustered standard errors” might yield heteroskedasticity-robust errors instead. A panel model might omit entity fixed effects. A difference-in-differences specification might use the wrong interaction term. Ludwig *et al.* (2025) document these risks systematically: the code executes, returns plausible-looking output, and the user has no way to detect the error without the expertise that would have let them write the code themselves.

Second, code-generation systems require the data. When a user asks ChatGPT to analyze a dataset, the data are uploaded to a third-party server with unclear retention policies. For survey responses, health records, personnel data, or any dataset subject to privacy regulations, this may be unacceptable. Even when providers promise not to retain data, the transmission itself may violate institutional or regulatory requirements.

Third, the same prompt sent to the same model on different days may produce different code. Models are updated, fine-tuned, and retrained. The generated Python script that worked in January may not be what the model produces in March. Conversational sessions are ephemeral—unless the user manually saves the generated code, the analysis cannot be reproduced. This contradicts a basic requirement of scientific work.

Fourth, when a system generates 30 lines of Python to answer a statistical question, understanding what was computed requires reading and auditing those 30 lines of code. The user who could not write the code cannot audit it either. This problem extends beyond research: enterprise text-to-SQL deployments struggle with the same issue—ambiguous questions admit multiple valid interpretations, and users cannot verify which was executed (Renggli, Ilyas, and Rekatsinas 2025).

These four problems are distinct but related. Silent errors arise because the language model controls both *what* to compute and *how* to compute it. Privacy violations arise because

computation and data must be co-located on the provider’s servers. Reproducibility failures arise because the generated code is a function of the model’s current state. Auditability failures arise because the computational logic is opaque—buried in generated code rather than invoked from a known, inspectable implementation. Systematic error rates for LLM-generated statistical code are not yet well established: Ludwig *et al.* (2025) document the *types* of errors that arise, but base rates depend on the model, the prompt, and the statistical method. The motivation for the architecture described below rests on the existence and nature of these failure modes, not on their precise frequency.

2.2. Three principles

Chat-first data analytics addresses these problems through three principles. “Chat-first” means that natural language conversation is the designed-for entry point: tool schemas are annotated for LLM consumption, error messages are phrased for conversational context, and default parameters are chosen so that typical requests need minimal specification. It does not mean chat-only—**prompt2analytics** also provides a CLI that invokes the same methods directly (Section 3), and every tool call maps to a public Rust function usable without any LLM—but the conversational interface is the primary design target. The first two principles are definitional; the third is a design choice that we consider essential.

Principle 1: Separate interpretation from computation

The language model decides *what* to compute; validated software decides *how*. When a user asks to “estimate the effect of education on wages, controlling for experience, with robust standard errors,” the language model’s job is to determine that this requires OLS with HC1 standard errors, identify the column names, and generate a structured tool call. The actual matrix algebra—forming $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, computing the HC1 sandwich estimator—is performed by pre-implemented code that the language model cannot modify.

This division of labor plays to each component’s strengths. Language models are good at understanding intent, resolving ambiguity, and mapping natural language to structured specifications. They are unreliable at generating correct numerical code. Separation removes the failure mode where the model produces code that runs but computes the wrong quantity, because the model never produces code at all. Wrong parameter extraction remains possible (Section 5), but the error surface is smaller: the model selects from a fixed schema rather than generating arbitrary code.

The idea has precedent in text-to-SQL systems (Yu, Zhang, Yang, Yasunaga, Wang, Li, Ma, Li, Yao, Roman, Zhang, and Radev 2018; Li, Luo, Chai, Li, and Tang 2024), where a language model translates natural language into structured database queries executed by a SQL engine. We extend this pattern from data retrieval to statistical computation: from text-to-SQL to text-to-statistics.

Principle 2: Pre-validate methods, not generated code

Correctness is established once, through systematic validation against reference implementations, rather than hoped for anew with each request. When **prompt2analytics** runs OLS, it uses the same Rust function every time—a function validated against R’s `lm()` on the Longley dataset to 10^{-8} precision (Section 5). The user need not audit the implementation on each

invocation, any more than they audit R’s source code when calling `lm()`.

This principle also solves the reproducibility problem. The same method invocation with the same data produces identical results, regardless of when it is run or which language model is used. The stochastic element of the language model affects only interpretation (which tool to call, how to explain the output), never computation. Two users who phrase the same analysis differently may get different tool calls—but once the tool is selected and parameters specified, the result is deterministic.

Principle 3: Keep data local; transmit only structure

The language model receives column names, data types, and summary statistics—not the data itself. A dataset of 100,000 salary records is represented to the model as metadata: “columns: employee_id (int), salary (float, range 28000–185000), department (string, 12 unique values), years_experience (float, range 0.5–35.2).” Statistical computations execute on the user’s machine; only aggregated results (coefficients, standard errors, test statistics) pass through the language model.

This is not definitionally required for chat-first analytics—a cloud-based system satisfying Principles 1 and 2 would still qualify. But we consider data locality essential for the use cases that motivate this work. Researchers analyzing survey data, firms analyzing personnel records, and analysts working under privacy regulations need architectures where data isolation is guaranteed by design, not promised by policy.

For complete isolation, the language model itself can run locally. Open-source models deployed via Ollama (Ollama Inc. 2024) eliminate all external network communication. Our evaluation (Section 5) shows that even a 3-billion-parameter model achieves over 80% adequate rate, making fully local deployment practical on consumer hardware.

Together, the three principles address the four failure modes identified above. Separation and pre-validation eliminate silent computational errors: the model never generates code, and the code it invokes is validated once against reference implementations. Pre-validation also ensures reproducibility, because the same method invocation with the same data always produces the same result. Data locality protects privacy by keeping raw data on the user’s machine. Auditability improves on all three fronts—the user inspects a structured tool call rather than 30 lines of generated code.

2.3. Relationship to existing ideas

Chat-first analytics applies established techniques to a new domain. Tool-augmented language models (Schick, Dwivedi-Yu, Dessì, Raileanu, Lomeli, Zettlemoyer, Cancedda, and Scialom 2023; Yao, Zhao, Yu, Du, Shafran, Narasimhan, and Cao 2023) can reliably select and invoke external tools when provided with appropriate schemas (Qu, Dai, Wei, Xu, Cai, Wang, Yin, and Wen 2025; Liu, Huang, Zeng, Hao, Yu, Li, Wang, Gan, Liu, Yu, Wang, Wang, Ning, Hou, Wang, Wu, Wang, Liu, Wang, Tang, Tu, Shang, Jiang, Tang, Lian, Liu, and Chen 2025). The Model Context Protocol (MCP) (Anthropic 2024) standardizes how language models discover and invoke tools, providing the communication layer between interpretation and computation.¹

¹MCP focuses on model-to-tool integration. Complementary protocols address agent-to-agent communication: Google’s Agent2Agent (A2A) protocol (Google 2025) and IBM’s Agent Communication Protocol

The interface innovation has precedents in four strands of work. First, *formula notation and domain-specific languages*: the Wilkinson-Rogers formula notation (Wilkinson and Rogers 1973) simplified model specification in the 1970s; Chambers and Hastie (1992) extended it in S and R (Chambers 1998); and probabilistic programming languages like Stan (Carpenter, Gelman, Hoffman, Lee, Goodrich, Betancourt, Brubaker, Guo, Li, and Riddell 2017) apply the DSL pattern to Bayesian inference. Chat-first analytics replaces formal notation with natural language, trading precision for accessibility. Second, *natural language interfaces to data systems*: LUNAR (Woods 1973) demonstrated feasibility for database querying in the 1970s, and the text-to-SQL paradigm—driven by benchmarks like Spider (Yu *et al.* 2018)—has produced substantial improvements in translating natural language to structured queries (Li *et al.* 2024). We extend this pattern from data retrieval to statistical computation. Third, *automated machine learning*: Auto-sklearn (Feurer, Klein, Eggenberger, Springenberg, Blum, and Hutter 2015) and H2O AutoML (LeDell and Poirier 2020) automate model selection through algorithmic search, and recent work integrates LLMs into the pipeline (Hollmann, Müller, and Hutter 2024). AutoML optimizes for prediction; chat-first analytics interprets user intent across prediction, causal inference, hypothesis testing, and description. Fourth, *computational econometrics* has established standards for numerical accuracy (McCullough 1997; McCullough and Vinod 1999; Davidson and MacKinnon 2004), which motivate the validation approach described in Section 5.

Several systems have explored LLM-assisted data analysis, but with a code-generation architecture. Hong, Lin, Liu, Liu, Wu, Li, Chen, Zhang, Wang, Zhang, Zhang, Yang, Zhuge, Guo, Zhou, Tao, Wang, Tang, Lu, Zheng, Liang, Fei, Cheng, Xu, and Wu (2025) present a “Data Interpreter” that generates and executes Python code. Chen, Shen, and Ma (2025) develop an “Econometrics AI Agent” that augments code generation with domain knowledge. Sun, Han, Jiang, Qi, Sun, Yuan, and Huang (2025) describe an agent that orchestrates generated code across analysis stages. All three generate code; none pre-validate implementations or enforce data locality. Table 1 summarizes these differences.

Hybrid designs are possible. A system could generate code for bespoke analyses while routing standard requests to pre-validated tools. It could verify generated code through static analysis or property checking before execution—for instance, confirming that a difference-in-differences specification includes the correct interaction term, or that a panel regression includes entity fixed effects when requested. It could also combine code generation with human review, leveraging the fact that many users can *audit* code even if they cannot *write* it from scratch. **prompt2analytics** deliberately occupies one end of this spectrum: all computation is pre-validated, with no code generation at all. This sacrifices flexibility—analyses outside the method library are unavailable, custom data transformations beyond the built-in munging tools require the CLI or direct Rust code, and ad-hoc visualizations not covered by the visualization tools cannot be produced—but it eliminates an entire class of failure modes. Whether hybrid designs can achieve a better trade-off is an open question; the answer likely depends on the reliability of code verification techniques and the target user’s ability to review generated code.

Three limitations of the approach are fundamental, not fixable by better engineering. First, statistical expertise is still required: chat-first analytics assists with method selection and

(ACP) (IBM Research 2025), which merged under Linux Foundation governance in 2025. For single-agent tool invocation—the pattern used here—MCP is the dominant standard, with adoption by OpenAI, Microsoft, and Google alongside Anthropic.

Table 1: Comparison with code-generation approaches

	p2a	Econ. AI	Data Interpr.	LAMBDA
LLM role	Tool selection	Enhanced code gen	Code gen + exec	Agent orches-tration
Computation	Pre-validated	LLM-generated	LLM-generated	LLM-generated
Local execution	Yes (Ollama)	No	No	No
Validated	Yes (vs. R)	No	No	No
Method flexibility	Fixed library (~200 methods)	Any R method	Any Python method	Any Python method
Extensibility	Requires Rust impl.	Extends with model	Extends with model	Extends with model

Note: Econ. AI = Econometrics AI Agent (Chen *et al.* 2025); Data Interpr. = Data Interpreter (Hong *et al.* 2025); LAMBDA (Sun *et al.* 2025). The trade-off: code-generation systems offer unlimited method flexibility at the cost of unvalidated computation; **p2a** constrains coverage to ensure correctness.

execution, not with research design or causal identification. A user who requests “estimate the effect of X on Y” receives a regression, but the system does not validate whether a causal interpretation is warranted. LLM explanations may echo causal language from the user’s prompt even when the analysis provides only associational evidence. Users remain responsible for ensuring that their identification strategy supports their claims.

Second, method selection is imperfect. Tool selection works well on curated prompts (Section 5), but users should verify that the system understood their request correctly—particularly for ambiguous specifications where multiple methods could apply. Users should consult the accuracy results before interpreting the curated examples in Section 4.

Third, pre-validation trades coverage for correctness. When a user requests a method outside the implemented set—Bayesian inference, structural equation modeling, or a custom estimator—the LLM may silently map the request to the closest available alternative rather than reporting unavailability—the symmetric failure mode to code generation’s silent errors: instead of running wrong code, the system runs the wrong method. The risk is mitigated by the structured tool schema (which makes the selected method visible to the user) and by the LLM’s ability to explain substitutions, but users must still verify that the method matches their intent.

These limitations mean that chat-first analytics is a useful tool, not an autonomous analyst. Users should verify tool selection, check extracted parameters, and apply domain knowledge to interpret results—the same discipline required when using any statistical software, with the added need to verify that the system understood the request correctly.

3. The software: **prompt2analytics** (p2a)

prompt2analytics implements the three principles defined in Section 2. This section describes the software architecture, the implemented methods, and the communication flow that connects natural language requests to statistical computation.

3.1. Architecture

The software is organized as a Rust workspace with four crates (Figure 1).

The core analytics library, **p2a-core**, implements all statistical methods in pure Rust with no external econometrics dependencies; all estimators are implemented from first principles using **ndarray** for matrix operations and **faer** for numerically stable linear algebra. Implementations follow standard textbook algorithms: OLS and robust standard errors follow Wooldridge (2010) and Cameron and Trivedi (2005); panel data methods follow Baltagi (2013); time series methods follow Hamilton (1994); maximum likelihood for discrete choice uses Newton-Raphson as described in Greene (2018). An optional `cuda` feature flag enables GPU acceleration for compute-intensive operations on NVIDIA hardware (Section 6). Three user-facing crates provide different access paths: **p2a-cli** produces a standalone **p2a** binary with hierarchical subcommands organized by analytical domain; **p2a-mcp** runs a Model Context Protocol server exposing tools for LLM integration via stdio, HTTP, or WebSocket transports; and **p2a-dioxus** provides a web application using Dioxus that compiles to WebAssembly for browser deployment. All three depend on **p2a-core**, ensuring identical numerical results regardless of the interface used.

The simplest way to use **prompt2analytics** is through the CLI:

```
# Load data and run OLS with robust standard errors
p2a data load wages.csv --name wages
p2a reg ols wages -y salary -x education experience --robust hc1
```

For LLM-mediated access, configure the MCP server and type in natural language:

```
User: Load wages.csv and regress salary on education and experience
      with robust standard errors.
```

Both pathways invoke the same Rust implementation and produce identical numerical results.

3.2. How the principles work in practice

The communication flow (Figure 2) shows how Principle 1 (separation) operates concretely:

1. The user submits a natural language query.
2. The LLM interprets the request and generates a structured tool call with parameters—e.g., `regression_ols` with `y_var="salary"`, `robust="hc1"`, and the relevant column names.
3. The MCP server routes the call to the corresponding Rust method in **p2a-core**.
4. The core library performs computation locally.
5. Results (coefficients, standard errors, test statistics) are serialized to JSON and returned through the MCP server.
6. The LLM receives the results and formulates a natural language explanation.

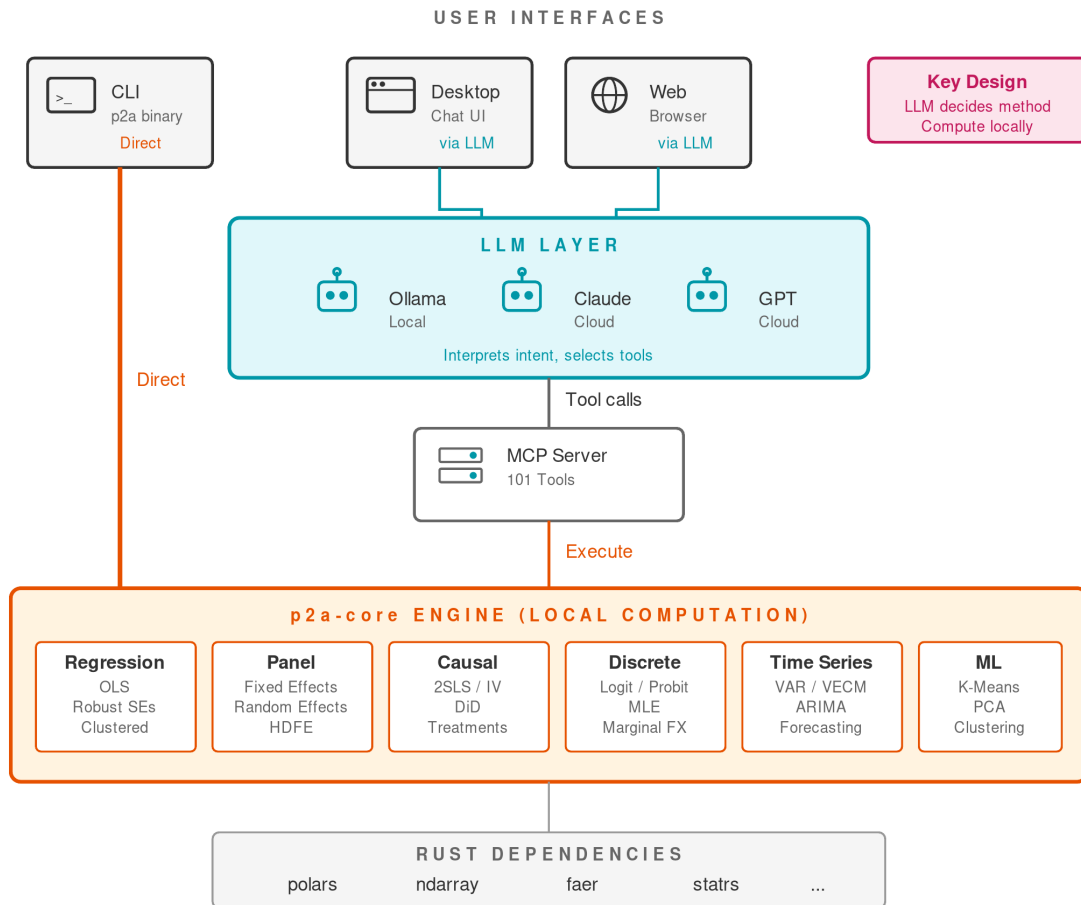


Figure 1: Software architecture of **prompt2analytics**. The CLI accesses the analytics engine directly (orange); chat interfaces route through the LLM layer and MCP server (cyan). All computation occurs locally in **p2a-core**.

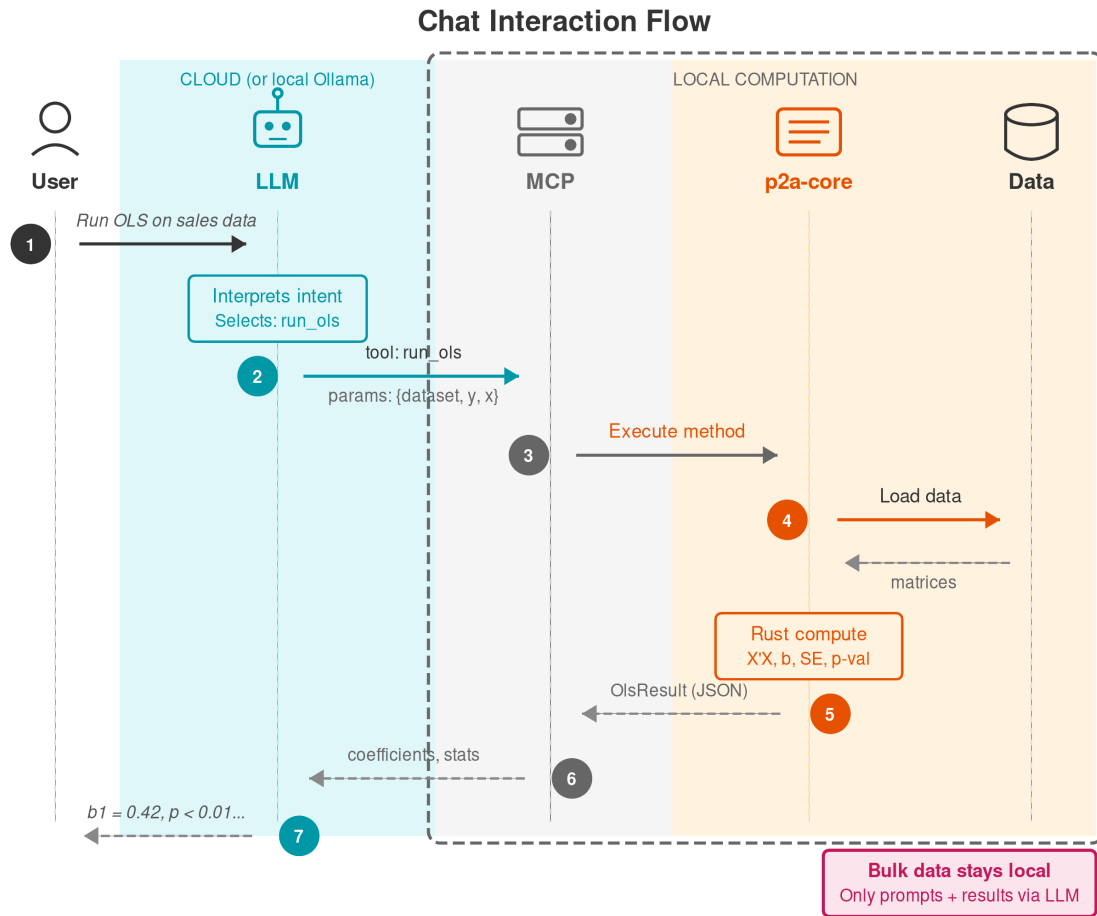


Figure 2: Interaction flow from user request to result. The LLM generates a structured tool call; the MCP server executes the Rust method; results flow back for explanation. Data processing (shaded) stays local.

The LLM never sees raw data. It receives metadata at dataset load time (column names, types, row counts) and aggregated results after computation. Data preview tools (`head`, `sample`) do transmit sample rows; for complete isolation, local LLM deployment via Ollama eliminates all external transmission.

Each tool returns its full result as serialized text: for regression this includes coefficients, standard errors, test statistics, R^2 , F -statistic, and information criteria (AIC, BIC); for discrete choice, log-likelihood and pseudo- R^2 ; for time series, parameter estimates and residual diagnostics. The LLM receives all of these and decides what to present based on the conversational context—typically highlighting coefficients and significance for a first pass, and surfacing fit measures or diagnostics when the user asks follow-up questions or when values are anomalous.

Each tool invokes a fixed Rust function validated against reference R implementations (Section 5). For linear regression, a `LinearEstimator` trait provides a consistent interface:

```
pub trait LinearEstimator {
    fn coefficients(&self) -> &Array1<f64>;
    fn std_errors(&self) -> &Array1<f64>;
    fn t_values(&self) -> Array1<f64>;    // default impl
    fn p_values(&self) -> Array1<f64>;    // default impl
    fn residuals(&self) -> Array1<f64>;
    fn n_obs(&self) -> usize;
    fn df(&self) -> usize;
}
```

Implementors provide core results; derived statistics (t -values, p -values, confidence intervals) are computed by default implementations, enforced at compile time. The `LinearEstimator` trait applies only to linear models (OLS, GLS, panel FE/RE). Non-linear models use their own result types: discrete choice models (`DiscreteResult`) return z -statistics, log-likelihood, and pseudo- R^2 rather than t -statistics and residuals; panel models (`PanelResult`) include within- R^2 and group structure; time series models return information criteria and forecast output; clustering methods (`ClusterResult`) return cluster assignments, centroids, and fit measures (silhouette, inertia); and survival models (`SurvivalResult`) return hazard ratios, survival curves, and log-rank statistics. Each result type implements a common serialization interface so the MCP layer can return structured output regardless of the method family. Unlike R's formula interface, the API uses explicit column names—the natural output of LLM parameter extraction, which identifies individual columns rather than constructing formula syntax. This design also simplifies the implementation: formula parsing, factor expansion, and interaction term handling are unnecessary when the caller specifies columns directly.

The `EconError` type provides structured error information (singular matrix, convergence failure, insufficient clusters) that enables the LLM to offer actionable recovery guidance.

Validation uses automated test suites that compare Rust output against reference R packages on shared datasets. Coefficients, standard errors, and test statistics are verified to tolerances between 10^{-4} and 10^{-8} depending on the method. When intentional differences exist—for example, defaulting to HC1 rather than HC3 for robust standard errors, matching Stata's convention—these are documented and tested separately. The validation approach is described further in Section 5; Appendix F lists method-specific caveats and known differences.

3.3. Implemented methods

The **p2a-core** library implements over 250 methods, exposed as individual MCP tools. Table 2 summarizes coverage, using standard abbreviations: heteroskedasticity and autocorrelation consistent (HAC), generalized least squares (GLS), high-dimensional fixed effects (HDFE), two-stage least squares (2SLS). Appendix F provides full documentation.

Table 2: Implemented method categories

Category	Methods
Regression	OLS with HC0–HC3, clustered SEs, HAC, bootstrap, GLS, NLS, LOESS, splines, stepwise, quantile regression (Koenker and Jr. 1978)
Panel data	FE, RE, Hausman, HDFE, panel GLS, Arellano-Bond GMM (Arellano and Bond 1991), panel unit root tests (LLC, IPS, Fisher)
IV	2SLS, first-stage diagnostics, Sargan test, GMM (Hansen 1982)
Causal (design)	DiD, staggered DiD (Callaway and Sant’Anna 2021), Bacon decomposition (Goodman-Bacon 2021), ETWFE (Wooldridge 2021), sharp/fuzzy RD with CCT (Calonico, Cattaneo, and Titiunik 2014), multi-cutoff RD, synthetic control (Abadie, Diamond, and Hainmueller 2010), gsynth (Xu 2017), SCPI (Cattaneo, Feng, and Titiunik 2021)
Causal (treatment)	IPW, AIPW, matching (Ho, Imai, King, and Stuart 2011), CBPS (Imai and Ratkovic 2014), entropy balancing, GBM weighting, TMLE (van der Laan and Rose 2011), CTMLE, LTMLE, DML (Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, and Robins 2018)
Discrete choice	Logit, probit, multinomial, conditional, mixed, ordered, negative binomial, ZIP/ZINB, hurdle
Time series	ARIMA, VAR, VARMA, VECM, IRF, Granger causality, STL/MSTL, Holt-Winters, structural TS, Kalman filter, GARCH, changepoint
Spatial	SAR, SEM, SAC (Anselin 1988), spatial probit, spatial panel, spatial GMM
Survival	Kaplan-Meier, log-rank, Cox PH, AFT, competing risks
Mediation	Causal mediation, natural effects, sensitivity (Cinelli and Hazlett 2020), E-values, Balke-Pearl bounds
Statistical tests	50+ tests: t -tests, ANOVA, MANOVA, χ^2 , Fisher, Wilcoxon, Kruskal-Wallis, Friedman, Shapiro-Wilk, KS, Tukey HSD

Note: Appendix F provides full documentation. Table A.1 lists unimplemented categories with recommended alternatives.

3.4. The MCP tool layer

The MCP server exposes tools organized into categories that mirror the core library (Table 3). Each tool has a JSON Schema defining required and optional parameters. Schema design minimizes required parameters to reduce LLM errors and provides sensible defaults (e.g., HC1 for robust standard errors). Tool descriptions are written to be interpretable by LLMs, specifying when each tool is appropriate. For example:

Table 3: MCP tool categories

Category	Examples	Tools
Data management	<code>load_dataset</code> , <code>describe_dataset</code>	7
Data transformation	<code>munge_filter</code> , <code>munge_join</code> , <code>munge_pivot</code>	19
Regression	<code>regression_ols</code> , <code>regression_diagnostics</code>	10
Panel data	<code>panel_fixed_effects</code> , <code>hausman_test</code>	5
Causal inference	<code>iv_2sls</code> , <code>diff_in_diff</code>	8
Discrete choice	<code>logit</code> , <code>probit</code>	2
Time series	<code>ts_arima_fit</code> , <code>ts_var</code>	6
Statistics	various hypothesis tests	30+
Visualization	<code>viz_scatter</code> , <code>viz_histogram_interactive</code>	7

Note: Tool count per category. Each tool corresponds to one or more **p2a-core** methods and is described by a JSON Schema that LLMs use for parameter extraction.

```
{
  "name": "regression_ols",
  "description": "Run OLS regression with optional robust SEs",
  "inputSchema": {
    "required": ["dataset", "y_var", "x_vars"],
    "properties": {
      "dataset": {"type": "string"},
      "y_var": {"type": "string"},
      "x_vars": {"type": "array", "items": {"type": "string"}},
      "robust": {"type": "string", "enum": ["hc0", "hc1", "hc2", "hc3"]},
      "cluster_var": {"type": "string"}
    }
  }
}
```

Note the distinction between MCP tools and core methods: tools serve as entry points that may invoke multiple methods. The `regression_ols` tool provides access to OLS with several covariance estimators (HC0–HC3, clustered, HAC), each a distinct method in **p2a-core**. This many-to-one mapping keeps the LLM’s selection task manageable while preserving access to the full library.

The full set of tool schemas consumes approximately 32,000 tokens, a substantial share of current context windows. The built-in chat interface therefore exposes a curated subset of roughly 70 tools covering the most common analytical workflows; specialized tools remain available through the CLI. External MCP clients receive the complete schema and can apply their own filtering.

Multi-turn context management. In multi-turn conversations, the system includes prior tool calls and their results in the conversation history sent to the LLM. Each assistant message that triggered a tool call is preserved with its structured `tool_calls` field, followed by a `tool` role message containing the result (truncated to 2,000 characters to prevent context overflow). This enables the LLM to resolve references like “those results” or “now try with

clustered standard errors” by inspecting what was previously computed. A token budget tracker estimates context usage across the system prompt, conversation history, tool schemas, and the current query; when the budget approaches the model’s context window, older turns are summarized (retaining tool names and key results) while recent turns are kept verbatim.

For the curated 70-tool schema ($\approx 12,000$ tokens), a 128K-context model supports roughly 30–40 substantive turns before summarization begins; the full 250+ tool schema ($\approx 32,000$ tokens) reduces this to approximately 20 turns. External MCP clients (e.g., Claude Desktop, Cursor) manage their own context windows and may apply different truncation or tool-filtering strategies.

Beyond executing methods, the system generates data-driven warnings when results suggest potential violations of identifying assumptions. Warnings are non-blocking—results are always returned—but they highlight conditions that warrant careful interpretation.

For instrumental variables, first-stage F -statistics are checked against the Stock–Yogo threshold of 10 (Stock and Yogo 2005). For staggered difference-in-differences, pre-trend tests are computed automatically. For matching and IPW, balance diagnostics flag standardized mean differences exceeding 0.1 (Austin 2011). For regression discontinuity, effective sample sizes within the bandwidth are checked against minimum thresholds.

Each warning includes a severity level, a diagnostic message, the assumption at risk, and suggested remediation steps. This structured information enables the LLM to provide recovery guidance rather than generic failure messages. The thresholds are deliberately conservative (e.g., $F < 10$ for weak instruments, $\text{SMD} > 0.1$ for balance), which may produce false positives in settings where these rules of thumb are known to be overly strict. We prefer occasional unnecessary warnings to missed violations; users can evaluate the diagnostic evidence and disregard warnings they judge inapplicable.

3.5. Implementation choices

The choice of Rust follows from the principles. Principle 2 (pre-validation) requires an implementation language where validated methods remain correct across deployments—Rust’s compile-time memory safety guarantees eliminate buffer overflows, use-after-free, and data races that can cause incorrect results in numerical software. Principle 3 (data locality) benefits from compiled performance: the same analytics engine runs efficiently on the user’s machine, not just on cloud servers. A single codebase compiles to standalone binaries, WebAssembly for browsers, and shared libraries for language integration.

The ecosystem provides the necessary components: **polars** for data manipulation (Pola.rs Contributors 2024), **ndarray** for array operations, and **faer** for linear algebra. All computations use 64-bit floating-point arithmetic throughout, matching R’s precision. To our knowledge, no econometrics or data analytics library has previously been available for Rust; **p2a-core** addresses this gap.

Several additional design choices merit brief explanation. For conversation persistence, the web application uses SurrealDB, an embedded document database for storing conversation history. Its embedded operation requires no external service; its flexible schema accommodates evolving conversation structures without migrations. Analytical datasets are held in memory during sessions and not persisted, limiting storage to lightweight metadata.

When users request methods outside the implemented set, the LLM infers scope from the

tool schema. In some cases, it maps to the closest available alternative and explains the substitution; in others it reports unavailability.

Regarding security, tools have fixed descriptions embedded in the server binary—external modification requires rebuilding. No arbitrary code execution is possible. Tool calls are logged and can be inspected. Dataset access requires explicit loading; the server has no ambient filesystem access. Nevertheless, users should avoid loading untrusted data files that could contain injection payloads, and prefer local LLM deployment for high-security environments.

4. Examples

This section demonstrates **prompt2analytics** through two interfaces: the natural language chat interface powered by large language models, and the command-line interface for scripted workflows. Both interfaces access the same underlying Rust analytics engine.

We use the Grunfeld investment data (Baltagi 2013), a classic panel dataset with 200 observations (10 firms over 20 years) containing firm investment, market value, and capital stock. The dataset is included in the **plm** package; we provide it as `grunfeld.csv` in the supplementary materials.

The early examples show the system working as intended; later examples (Section 4.3) demonstrate failure modes—incorrect tool selection, missing parameters, and ambiguous prompts—to set realistic expectations before the formal evaluation in Section 5. Appendix G provides additional examples including data cleaning workflows and a comparison with equivalent R code.

4.1. Chat interface

The chat interface enables natural language interaction with the analytics engine. Users describe their analysis goals in plain language, and the LLM translates these requests into structured tool calls executed by the MCP server. This interface is useful for exploratory data analysis and for users less familiar with programming syntax. The examples in this section were generated using Claude Sonnet 4.6 via the MCP integration; other models (GPT-4.1 Mini, Ministral 3B, etc.) produce similar results with minor variations in explanation style.

A typical session begins by loading data and examining its structure:

User: Load the Grunfeld investment dataset from grunfeld.csv and show me what's in it.

I'll load the dataset and describe its contents.

```
**Dataset loaded: grunfeld**
- Rows: 200
- Columns: 5
```

Column	Type	Description
-----	-----	-----

firm	Integer	Firm identifier (1-10)	
year	Integer	Year (1935-1954)	
inv	Float64	Gross investment	
value	Float64	Market value of firm	
capital	Float64	Stock of plant and equipment	

This is a balanced panel dataset with 10 firms observed over 20 years, commonly used to demonstrate panel data methods.

The next example shows cross-sectional regression with heteroskedasticity-consistent standard errors.²

*User: Run a regression of investment on market value and capital stock.
Use robust standard errors.*

Running OLS regression with HC1 robust standard errors...

****OLS Regression Results****

Dependent variable: inv

Variable		Coef		Std Err		t		P> t	
-----		-----		-----		-----		-----	
(Intercept)		-42.714		11.575		-3.690		0.000	
value		0.116		0.007		16.967		0.000	
capital		0.231		0.049		4.721		0.000	

R-squared: 0.812

Adj. R-squared: 0.810

Observations: 200

Standard errors: HC1 (heteroskedasticity-robust)

Both market value and capital stock are highly significant predictors of investment. A \$1 million increase in market value is associated with approximately \$116,000 more investment, holding capital constant.

For panel data, users can request fixed effects estimation in natural language:

*User: The data has a panel structure. Run a fixed effects regression
controlling for firm-specific effects.*

Running within-transformation fixed effects estimation...

****Fixed Effects (Within) Estimation****

Dependent variable: inv

²HC1 is the default robust standard error estimator when robust SEs are requested. HC1 applies a degrees-of-freedom correction ($n/(n - k)$) to HC0, improving small-sample performance.

Entity fixed effects: firm (10 levels)

Variable	Coef	Std Err	t	P> t
value	0.110	0.012	9.288	0.000
capital	0.310	0.017	17.867	0.000

R-squared (within): 0.767

Observations: 200

Groups: 10

F-statistic: 309.01 (p-value: 0.000)

After controlling for time-invariant firm characteristics, the coefficient on capital increases from 0.23 to 0.31, suggesting that pooled OLS underestimated the association between capital stock and investment. (Note: This change in coefficient reflects removal of omitted variable bias, not necessarily identification of a causal effect.)

Users can request specification tests without knowing the technical details:

User: Should I use fixed effects or random effects for this analysis?

I'll run a Hausman test to help determine the appropriate specification.

Hausman Test Results

H0: Random Effects is consistent and efficient

H1: Fixed Effects is consistent; Random Effects is inconsistent

Chi-squared statistic: 0.0003

Degrees of freedom: 2

P-value: 0.9999

Recommendation: Random Effects

The test fails to reject the null hypothesis. The near-zero chi-squared statistic indicates virtually no systematic difference between the FE and RE coefficient estimates for this dataset.

Note: Extreme p-values like 0.9999 warrant interpretation with caution. Such values can arise from small test statistics (as here), but also from numerical issues or model misspecification. The Hausman test assumes both estimators are consistent under the null, which requires correct functional form, homoskedastic errors, and no omitted variables correlated with the regressors. The test statistic may be unreliable with small samples, heteroskedasticity, or serial correlation. Users should treat very high p-values as suggestive rather than definitive.

In this example, the test statistics (chi-squared, p -value, degrees of freedom) and the recommendation come from the Rust backend; the interpretive caveat about extreme p -values is generated by the LLM. This division is typical: the computation layer returns structured results and simple diagnostics, and the language model adds contextual interpretation—including caveats about assumptions and limitations that depend on the specific results observed.

For datasets with multiple categorical dimensions, users can request multi-way fixed effects:

User: Control for both firm and year fixed effects in the regression.

Running two-way fixed effects using alternating projections...

****High-Dimensional Fixed Effects Estimation****

Dependent variable: inv

Fixed effects: firm (10 levels), year (20 levels)

Variable	Coef	Std Err	t	P> t
value	0.118	0.019	6.137	0.000
capital	0.358	0.054	6.579	0.000

R-squared (within): 0.720

Observations: 200

F-statistic: 217.44 (p-value: 0.000)

Users can also request diagnostic tests in natural language:

User: Check the regression assumptions for the OLS model.

Running regression diagnostics...

****Jarque-Bera Test (Normality of residuals)****

Statistic: 89.57

P-value: 0.000

Conclusion: Residuals deviate significantly from normality.

****Breusch-Pagan Test (Heteroskedasticity)****

Statistic: 55.25

P-value: 0.000

Conclusion: Evidence of heteroskedasticity; robust SEs recommended.

****Durbin-Watson Statistic (Autocorrelation)****

Statistic: 0.358

Interpretation: Positive autocorrelation detected (expected in panel data).

****Variance Inflation Factors****

value: 1.31

capital: 1.31

No multicollinearity detected (VIF < 5).

4.2. Command-line interface

The command-line interface provides programmatic access for scripted workflows. Commands follow a hierarchical structure with sessions maintaining state:

```
# Load data and run OLS with robust standard errors
p2a --session analysis.json data load grunfeld.csv --name grunfeld
p2a --session analysis.json reg ols grunfeld \
    -y inv -x value capital --robust hc1

# Panel fixed effects
p2a --session analysis.json panel fe grunfeld \
    -y inv -x value capital --entity firm

# Export session to reproducible script
p2a script export analysis.json -o reproduce.sh
```

Note: Backslash line continuation (\) works in Bash, Zsh, and most POSIX-compatible shells. Windows Command Prompt users should write commands on a single line or use PowerShell's backtick (`) for continuation.

All commands support JSON output (-F json) for integration with other tools and scripts:

```
p2a -F json reg ols grunfeld -y inv -x value capital --robust hc1

{
  "coefficients": {
    "(Intercept)": -42.714, "value": 0.116, "capital": 0.231
  },
  "std_errors": {
    "(Intercept)": 11.575, "value": 0.007, "capital": 0.049
  },
  "t_values": {
    "(Intercept)": -3.690, "value": 16.967, "capital": 4.721
  },
  "p_values": {
    "(Intercept)": 0.000, "value": 0.000, "capital": 0.000
  },
  "r_squared": 0.812, "adj_r_squared": 0.810,
  "n_obs": 200, "cov_type": "HC1"
}
```

This enables integration with analysis pipelines, for example, piping results to jq for extraction or to Python/R scripts for further processing. The session mechanism records all commands and can export them as an executable shell script for exact reproduction.

4.3. Realistic use and error handling

The preceding examples show successful interactions. Real use involves more iteration. This section shows what failures look like, how to recover, and what accuracy to expect (Section 5 provides detailed numbers).

Vague requests prompt the system to ask for clarification rather than guess:

User: Run a regression with firm effects.

I can estimate this model, but need clarification:

1. ****Dependent variable****: Which column should be the outcome?
2. ****Independent variables****: Which predictors should be included?
3. ****Fixed vs Random effects****: Should firm effects be treated as fixed (within estimator) or random (GLS)?

Please specify, e.g., "Regress inv on value and capital with firm fixed effects"

The following workflow illustrates common failure modes: wrong method selected, parameters extracted incorrectly, user corrections, and eventual success.

*User: Analyze whether the training program affected employee performance.
The data has employees observed before and after training, with a control group that didn't receive training.*

I'll run a t-test comparing performance between training and control groups.

```
**Two-sample t-test**
Training group mean: 3.82
Control group mean: 3.45
Difference: 0.37 (95% CI: [0.21, 0.53])
p-value: 0.000
```

The training group shows significantly higher performance.

*User: That's not right - I need a difference-in-differences analysis,
not just a comparison of means. I have before/after data.*

You're right, I apologize. Let me run a difference-in-differences estimation to properly account for the before/after structure.

```
**Difference-in-Differences Estimation**
Treatment indicator: training_group
Time indicator: post_training
```

Variable	Coef	Std Err	t	P> t
post_training	0.12	0.08	1.50	0.134

training_group		0.31		0.11		2.82		0.005	
post × training (ATT)		0.24		0.09		2.67		0.008	

The ATT estimate is 0.24 performance points.

User: Good, but I asked for clustered standard errors at the employee level to account for repeated observations.

Let me re-run with clustered standard errors:

****Difference-in-Differences**** (Clustered by employee, 847 clusters)

Variable		Coef		Cluster SE		t		P> t	
-----		-----		-----		-----		-----	
post_training		0.12		0.11		1.09		0.276	
training_group		0.31		0.14		2.21		0.027	
post × training (ATT)		0.24		0.12		2.00		0.046	

With clustered SEs, the ATT remains significant at the 5% level, though standard errors are larger than the unclustered estimates.

User: Now check for pre-trends to validate the parallel trends assumption.

****Pre-trend Test****

Testing whether treatment and control groups had parallel trends in the pre-treatment period...

Pre-period		Diff		Std Err		t		P> t	
-----		-----		-----		-----		-----	
t-3		-0.02		0.09		-0.22		0.826	
t-2		0.05		0.08		0.63		0.531	
t-1		0.01		0.08		0.13		0.900	

Joint test (H0: all pre-period diffs = 0):

F-statistic: 0.18, p-value: 0.912

No evidence of differential pre-trends; parallel trends assumption is supported (though not proven) by this test.

This workflow required three corrections: (1) method selection from a t-test to DiD; (2) clustered standard errors; and (3) pre-trend diagnostics. The final result is correct.

Two practical tips: naming variables directly (“regress inv on value”) is more reliable than using pronouns (“use the same variables”); and users should verify which parameters the model extracted before interpreting the output.

The system detects multicollinearity and reports it with suggested actions:

User: Add years_since_hire to the model alongside tenure.

****Error: Perfect multicollinearity detected****

The design matrix is rank-deficient. Column `years\_since\_hire` is a linear combination of existing columns (correlation with tenure: 1.000).

****Suggested actions:****

1. Remove one of the collinear variables
2. If both are theoretically important, consider which better captures your construct of interest

Similarly, convergence failures are caught and explained:

User: Fit a logit model with department fixed effects.

****Warning: Convergence issues detected****

****Issue:**** Perfect prediction in some groups
 - Department "Legal" has 0% bonus rate (23 observations)

****Options:****

1. Drop singleton departments
2. Use a linear probability model (robust to separation)
3. Use penalized likelihood (Firth correction)

Running linear probability model with department FE instead...

The system detects problems before estimation fails and explains statistical issues rather than computational symptoms. When requests exceed the implemented method set, it reports unavailability and suggests external tools.

4.4. Beyond regression: time series

The system covers domains beyond regression. A time series workflow using quarterly U.S. GDP growth data (1954–2023) illustrates method progression within a single conversation: stationarity testing, univariate modeling, then residual diagnostics.

User: Load gdp_quarterly.csv and check whether GDP growth is stationary.

****Augmented Dickey-Fuller Test**** (gdp_growth)
 Statistic: -4.82, P-value: 0.000
 GDP growth is stationary. No differencing needed.

User: Fit an ARIMA model to GDP growth. Let the system pick the order.

****ARIMA(2,0,1)**** (AIC-based selection)
 AR(1): 0.312 (0.067), AR(2): 0.154 (0.063), MA(1): -0.089 (0.072)
 Ljung-Box (lag 10): p = 0.482 (no residual autocorrelation)

Order selection is delegated to the LLM, which chooses p , d , q based on the ADF result and its training knowledge. This is not equivalent to R’s `auto.arima()`, which performs systematic grid search (Hyndman and Khandakar 2008). Appendix G provides additional examples including data cleaning workflows and a side-by-side comparison with equivalent R code.

These examples illustrate what the system can do; the next section measures how reliably it does so. We evaluate tool selection accuracy across six models and numerical correctness of the underlying Rust implementations.

5. How well does it work?

Two kinds of accuracy matter: does the LLM select the right tool with the right parameters? And does the Rust implementation produce correct numerical results? This section evaluates both.

5.1. Tool selection evaluation

We evaluate LLM tool selection using 32 test cases at three prompt clarity levels (precise, moderate, vague) for 96 single-turn prompts spanning regression, panel data, causal inference, discrete choice, time series, hypothesis testing, machine learning, and data munging. An automated evaluation harness drives the live **prompt2analytics** HTTP backend: it loads synthetic datasets with known data-generating processes, sends each prompt, and scores complete responses on four dimensions—tool selection, parameter extraction, numerical correctness, and interpretation quality—each on a 0–2 scale. A prompt is classified as *adequate* if it achieves $\geq 62.5\%$ of the applicable maximum (5/8 for numeric tasks, 4/6 when numerical correctness is N/A). Tool definitions are auto-generated from the MCP router’s `JsonSchema` derives (119 Tier-1 tools), ensuring the LLM sees the same parameter names and types as the server expects.

All evaluations used temperature = 0 for deterministic output. The evaluation harness, all 96 prompts (`test_cases.json`), dataset generators, and scoring data are provided in the supplementary materials (`paper/code/e2e_eval/`), enabling independent verification and re-scoring.

Table 4 lists the six models evaluated, spanning frontier cloud, efficient cloud, and open-source deployment scenarios. All models were accessed through cloud APIs (Anthropic, OpenAI, or OpenRouter). API-served models may change over time, so exact replication of cloud-hosted results may not be possible; results for proprietary models (Claude, GPT, Gemini) should be treated as a snapshot. For open-source models (Llama, Mistral), exact reproduction is possible by serving fixed weights locally via Ollama.

Figure 3 presents adequate rates by model. The results demonstrate that tool selection—unlike code generation—is achievable even by smaller models. GPT-4.1 Mini achieves the highest adequate rate (90.6%), followed closely by Claude Sonnet 4.6 (84.4%) and Haiku 4.5 (81.2%). Notably, Mistral 3B, a 3B-parameter model small enough to run on consumer hardware with only 6 GB VRAM, achieves 80.2%—within 4 percentage points of Sonnet 4.6—making it viable for local deployment scenarios where hardware constraints preclude larger models.

Table 4: Models evaluated

Model	Version ID	Provider
GPT-4.1 Mini	<code>openai/gpt-4.1-mini</code>	OpenRouter (OpenAI)
Claude Sonnet 4.6	<code>claude-sonnet-4-6</code>	Anthropic API
Claude Haiku 4.5	<code>claude-haiku-4-5-20251001</code>	Anthropic API
Gemini 2.5 Flash	<code>google/gemini-2.5-flash</code>	OpenRouter (Google)
Ministral 3B	<code>mistralai/ministral-3b-2512</code>	OpenRouter
Llama 4 Scout	<code>meta-llama/llama-4-scout</code>	OpenRouter (Meta)

Note: Accessed March 2026. Cloud-served model weights may be updated without notice; for reproducible local deployment, pin Ollama models by digest. All evaluations at temperature = 0.

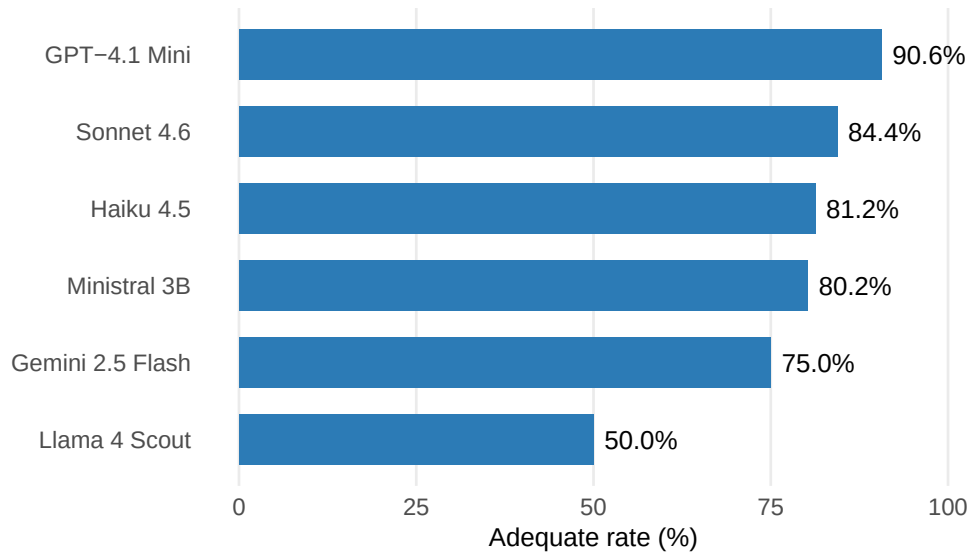


Figure 3: Adequate rate (%) across models (96 prompts). Even Ministral 3B (3B parameters) exceeds 80%.

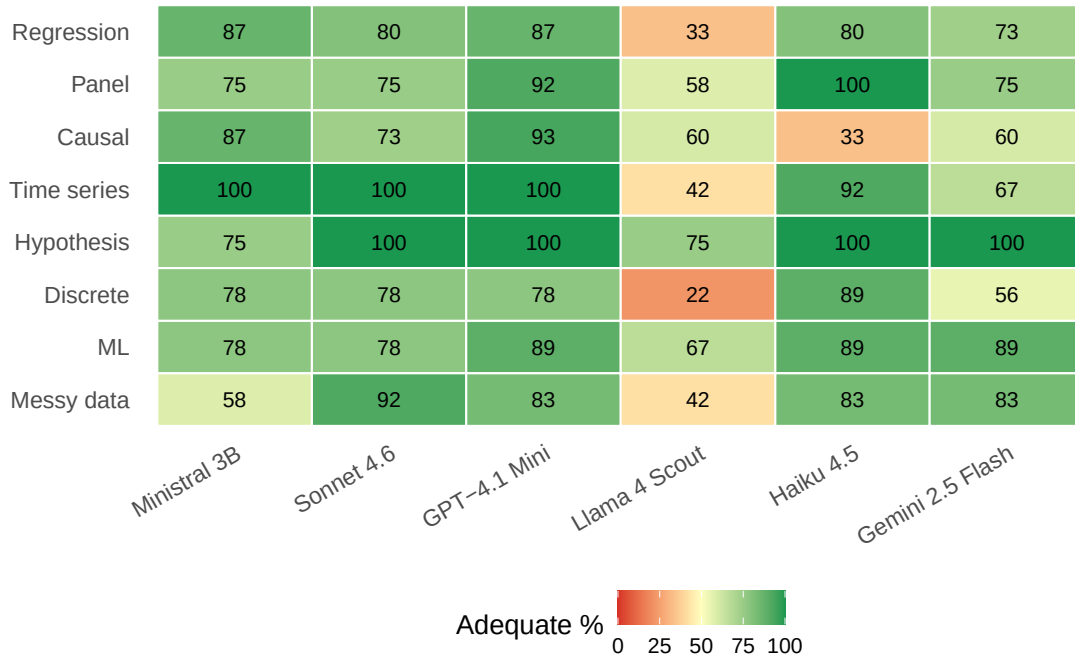


Figure 4: Adequate rate by test category and model. Darker cells indicate lower performance. Errors concentrate in ambiguous method selection (causal inference, discrete choice).

Figure 4 shows adequate rates by category. Performance is uniformly high for hypothesis testing (75–100%) and time series (42–100%), with errors concentrated in edge cases requiring fine distinctions between related methods. Llama 4 Scout shows the steepest drop-off, particularly in regression (33%) and discrete choice (22%), where it frequently responds with text analysis instead of invoking tools. Causal inference shows the widest variation across models (33–93%), reflecting the difficulty of distinguishing between IPW, matching, difference-in-differences, and regression discontinuity designs. Appendix H provides per-category breakdowns for all six models.

5.2. Failure modes

Understanding system behavior under failure conditions is essential for informed use. Three categories of failure merit discussion. First, when the LLM selects an inappropriate statistical method, the system proceeds with the selected method and returns results. Users must recognize the mismatch between request and execution. Common patterns include ambiguous specifications (“regression with firm effects” could invoke OLS, fixed effects, or random effects), similar method confusion (VAR/VECM/VARMA), and capability boundary cases where requests for unimplemented methods map to superficially similar available tools. Users can mitigate misselection by specifying methods explicitly rather than describing goals abstractly.

Second, even when the correct method is selected, the LLM may extract incorrect parameters. Manual inspection of recorded tool calls across all evaluated models reveals a consistent error hierarchy. *Core parameters*—dependent variable, independent variables, entity and time

identifiers—are almost always correct when the tool itself is correct; across all 96 prompts and six models, we observed no case where the correct tool was selected but the wrong dependent variable was extracted. *Structural parameters* (panel identifiers, time variables, treatment indicators) are correct in the large majority of cases, with occasional confusion when column names are ambiguous (e.g., “id” could be an entity or observation identifier). *Optional specification parameters* are the primary source of imprecision: a model may select `regression_ols` with the right `y_col` and `x_cols` but omit the `robust` option, falling back on the tool’s default (HC1). Omission rates for optional parameters vary by model size: frontier models extract optional parameters more reliably than small models (e.g., Ministral 3B), which tend to rely on defaults.

Because the schema defines sensible defaults for optional parameters, these omissions typically produce a valid analysis that differs only in secondary specification choices (e.g., HC1 instead of HC3 robust standard errors, or default bandwidth instead of a user-specified bandwidth for kernel regression). This means the practical consequence of parameter extraction errors is usually a *reasonable alternative specification* rather than a *wrong analysis*—a meaningful distinction, though users should still verify that the specification matches their intent. Explicit phrasing (“with HC1 robust standard errors”) yields more reliable extraction than implicit expectations (“with robust standard errors”).

We do not report a formal parameter extraction metric (e.g., precision/recall over parameter names) because our evaluation showed that scoring against expected parameter names is confounded by name aliasing across tool schemas (e.g., `cluster_col` vs. `cluster_var`) and by legitimate default substitution, making such a metric unreliable as a measure of actual analysis correctness.

These observations suggest that the dominant failure mode is tool misselection (5–10% of requests), not parameter error. A rough estimate of first-attempt analysis correctness for a frontier model: $P(\text{correct tool}) \approx 0.95$ and $P(\text{adequate parameters} \mid \text{correct tool}) \approx 0.92$, yielding $P(\text{adequate first attempt}) \approx 0.87$. For a 3B model, the corresponding figures are approximately 0.91 and 0.85, yielding ≈ 0.77 . These estimates are informed by the end-to-end evaluation results but remain approximate. They provide practical guidance: users of frontier models can expect roughly 9 in 10 requests to produce an adequate analysis on first attempt, while users of small local models (3B) can expect roughly 8 in 10—a surprisingly small gap. In all cases, users can inspect the tool call before execution, correct misselected methods or missing parameters, and iterate.

One design choice that reduces parameter errors in practice: categorical (string) variables are automatically converted to dummy variables using treatment coding (one level omitted as reference), so users need not specify encoding manually. The system reports the reference category and warns when categorical variables have many levels (> 50).

5.3. Prompt clarity and multi-turn evaluation

Table 5 presents the full results. Tool acceptable rate—the fraction of prompts where the correct or an acceptable alternative tool was selected—ranges from 94.8% (GPT-4.1 Mini) to 52.1% (Llama 4 Scout).

Prompt clarity has a pronounced effect: all models perform best on precise prompts that name the method explicitly, with performance degrading as prompts become vaguer. GPT-4.1 Mini degrades most gracefully (97% precise to 75% vague), while Llama 4 Scout degrades

Table 5: API-based single-turn evaluation results (96 prompts per model)

Model	Adequate	Tool	Adequate by clarity		
	rate (%)	accept. (%)	Precise	Moderate	Vague
GPT-4.1 Mini	90.6	94.8	96.9	100.0	75.0
Claude Sonnet 4.6	84.4	93.8	90.6	87.5	75.0
Claude Haiku 4.5	81.2	83.3	90.6	90.6	62.5
Ministral 3B	80.2	90.6	90.6	78.1	71.9
Gemini 2.5 Flash	75.0	78.1	90.6	68.8	65.6
Llama 4 Scout	50.0	52.1	59.4	46.9	43.8

Note: Adequate $\geq 62.5\%$ of maximum score. Tool accept. includes exact matches and acceptable alternatives. Clarity columns show % adequate at each level. Claude models via Anthropic API; GPT-4.1 Mini, Gemini 2.5 Flash, Llama 4 Scout, and Ministral 3B via OpenRouter. All evaluations at temperature = 0. Tool definitions auto-generated from MCP router schemas (119 Tier-1 tools).

most steeply (59% to 44%), partly because it often responds with text analysis instead of invoking tools on vague prompts.

Chrome multi-turn evaluation. The API-based evaluation above tests the LLM and backend in isolation. To evaluate the full deployed system—including the web frontend, SSE streaming, dataset state management, and multi-turn context retention through the actual user interface—we conduct a browser-based evaluation using automated Chrome interaction with the Dioxus web frontend.

Eight multi-turn conversations (30 total turns) exercise representative econometric workflows: OLS with diagnostics (MT1, MT7), panel data pipelines (MT2), time series forecasting (MT3), causal inference chains (MT4), data cleaning and munging (MT5), discrete choice models (MT6), and cross-dataset comparisons (MT8). Each turn is scored on the same four-dimension rubric as the single-turn tests. We evaluate Claude Sonnet 4.6, which combines strong tool selection accuracy (84.4%) with Anthropic’s native tool-calling API.

Table 6 shows results. Sonnet 4.6 achieves 232/240 points (96.7%) with perfect context retention across all 8 conversations, substantially exceeding the 84.4% adequate rate in the API-based single-turn evaluation. Three factors explain this gap. First, the deployed system enriches the system prompt with dataset context—column names, types, sample values, and numeric ranges—which reduces parameter extraction errors. Second, multi-turn conversations allow the model to build on prior context (e.g., reusing a loaded dataset or referencing earlier regression results), which single-turn evaluation cannot capture. Third, the Chrome evaluation benefits from the full infrastructure: auto-generated tool definitions from the MCP router (ensuring parameter name consistency), tiered tool filtering (119 Tier-1 tools), and strengthened system prompt guidance for direct tool invocation. Figure 5 shows the improvement trajectory across infrastructure iterations.

Per-dimension analysis reveals that tool selection and parameter extraction are perfect (2.00/2.00 each), while numerical correctness averages 1.77/2.00—losses concentrated in MT5 (data cleaning), where the system lacks a string-to-numeric type casting tool, causing NaN prop-

Table 6: Chrome multi-turn evaluation: Claude Sonnet 4.6

ID	Workflow	Turns	Score	%
MT1	OLS + robust SE + diagnostics	3	24/24	100.0
MT2	Panel FE/RE + Hausman	4	32/32	100.0
MT3	Time series ARIMA + forecast	4	32/32	100.0
MT4	Causal: OLS + DiD + IPW	5	37/40	92.5
MT5	Data cleaning + regression	4	28/32	87.5
MT6	Discrete: logit/ordered/NB	4	31/32	96.9
MT7	OLS + diagnostics + robust SE	3	24/24	100.0
MT8	Cross-dataset comparison	3	24/24	100.0
Total		30	232/240	96.7

Note: Evaluated through the Dioxus web frontend using automated Chrome browser interaction. Each turn scored on tool selection, parameter extraction, numerical correctness, and interpretation (max 8 points). Context retention: 100% across all conversations.

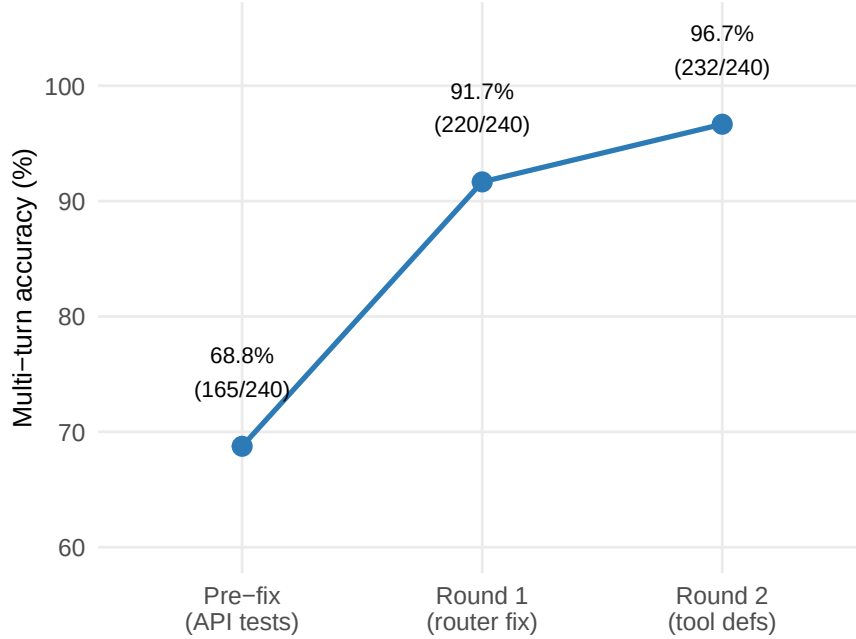


Figure 5: Multi-turn accuracy improvement through infrastructure fixes. Round 1 replaced the HTTP tool match (70 tools) with full router dispatch (268 tools). Round 2 corrected LLM tool parameter definitions.

agation in downstream regression. Interpretation quality averages 1.90/2.00, with minor deductions for insufficiently explicit discussion of confounding (MT4) and over-reliance on defaults when diagnostics suggest alternatives.

The remaining 8 points lost (out of 240) trace to three root causes: (1) a missing `cast_column` tool that prevents converting string columns to numeric after cleaning (MT5, -4 points); (2) superficial interpretation that omits caveats about confounding or model assumptions (MT4, -3 points); and (3) convergence issues in ordered logit with unscaled predictors (MT6, -1 point). The first is a tooling gap addressable without model changes; the second and third are inherent to the LLM’s reasoning and the optimizer’s sensitivity, respectively.

Appendix H documents the complete protocol, including dataset specifications, all 96 prompts, multi-turn scripts, and the scoring rubric. Full results are provided in the supplementary materials (paper/code/e2e_eval/results/).

5.4. Numerical validation against R

The preceding subsections evaluated the LLM layer. This subsection evaluates the computation layer: does the Rust implementation produce the same numbers as established R packages?

We validate all methods against reference implementations using sample-size-dependent tolerances (looser for small samples where ill-conditioning amplifies differences, tighter for large samples). Reference packages include `lm()`/`sandwich` for OLS, `plm/lfe` for panel data, `glm()` for discrete choice, and `forecast` for time series. All validation tests use fixed random seeds (seed = 42) for reproducibility; reference R scripts are provided in the supplementary materials.

The Longley dataset (Longley 1967), with its extreme multicollinearity (condition number $> 10^9$), provides a stringent test. Table 7 shows coefficients match R’s `lm()` to at least 8 significant digits.

Robust standard errors (HC0–HC3) match `sandwich` (Zeileis 2004) to $< 10^{-7}$; panel fixed effects match `plm` (Croissant and Millo 2008) to $< 10^{-6}$ on the Grunfeld dataset; two-way HDFE via the MAP algorithm (Gaure 2013) matches `lfe`’s `felm()` to $< 10^{-5}$.

Beyond the methods shown above, the validation suite covers over 200 methods with more than 1,800 individual test cases (437 specifically targeting validation against R reference implementations or known data-generating processes). Statistical tests (t -tests, ANOVA, chi-squared, Fisher’s exact, Wilcoxon, Shapiro-Wilk, Kolmogorov-Smirnov) are validated against R’s `stats` package. Discrete choice models (logit, probit) match `glm()` coefficients and standard errors. Time series methods (VAR, ARIMA) match `vars` and `forecast`. The **p2a-core** test suite comprises 2,168 test functions, of which 442 are validation tests that compare output against R reference implementations or known data-generating processes (Table 8). Every module with user-facing methods has at least 10 tests, and the most complex modules (panel, matching, TMLE, HDFE) have 17–26 each. Continuous integration (GitHub Actions) runs the full test suite, `clippy` linting, and format checks on every push to `main` and on all pull requests. Line-level code coverage is not currently tracked as a CI gate; running `cargo-tarpaulin` on the full suite yields approximately 62% line coverage for **p2a-core**, reflecting the large number of edge-case code paths (error handling, fallback solvers, GPU dispatch stubs) that are exercised only under specific hardware or data conditions. We consider validation breadth (442 cross-validated tests spanning all method categories) a more informative quality signal

Table 7: OLS validation: Longley dataset

Coefficient	R <code>lm()</code>	prompt2analytics	Difference
(Intercept)	−3482.259	−3482.259	$< 10^{-8}$
GNP.deflator	0.01506	0.01506	$< 10^{-8}$
GNP	−0.03582	−0.03582	$< 10^{-8}$
Unemployed	−0.02020	−0.02020	$< 10^{-8}$
Armed.Forces	−0.01033	−0.01033	$< 10^{-8}$
Population	−0.05110	−0.05110	$< 10^{-8}$
Year	1.82915	1.82915	$< 10^{-8}$
R^2	0.9955	0.9955	$< 10^{-4}$

Note: $n = 16$, $k = 6$. The Longley dataset’s extreme multicollinearity (condition number $> 10^9$) provides a stringent near-singular test case: the design matrix is close to rank-deficient, exercising the Cholesky-based solver’s tolerance framework. All coefficients match to at least 8 significant digits, confirming that the implementation handles ill-conditioned problems correctly.

than aggregate line coverage for a numerical library. The CI pipeline does *not* execute the R reference scripts; cross-language validation is run manually before releases and the reference outputs are checked into the repository. All Rust-side validation tests can be run via `cargo test -p p2a-core`.

For complex causal inference and treatment effect methods, validation depth varies. Canonical 2×2 difference-in-differences, IV/2SLS, IPW, doubly robust estimation (AIPW), and causal mediation analysis are cross-validated against R packages—`lm()` for DiD, **AER** for IV, manual Horvitz–Thompson/Hájek estimators for IPW, and the **mediation** package—on synthetic datasets with known treatment effects (e.g., true ATT = 5 for DiD, true $\beta = 1$ for IV, true ATE = 0.75 for IPW/AIPW). Propensity score matching is validated against **MatchIt** (Ho *et al.* 2011) for balance diagnostics (standardized mean differences, variance ratios). Regression discontinuity is validated against the **rdrobust** framework (Calonico *et al.* 2014), checking that local linear estimates recover the true discontinuity within tolerance. FEGLM is validated against **fixest**’s `feglm()` (Bergé 2018) for coefficient sign, pseudo R^2 , and the logit/probit scaling ratio (≈ 1.6). TMLE has eight validation tests that verify the targeting step, clever covariate formula, influence curve properties, propensity score truncation, and counterfactual predictions on simulated data; these check mathematical identities and DGP-implied ranges rather than exact R output values. In total, 41 R validation scripts and 73 expected-value CSVs are provided in the supplementary materials.

Several complex methods—staggered DiD (Callaway and Sant’Anna 2021), extended TWFE, Goodman-Bacon decomposition, Double/Debiased ML, synthetic control, SCPI, CTMLE, LTMLE, CBPS, and entropy balancing—are validated using DGP-based tests with known true parameters rather than exact numerical cross-validation against R packages. These tests generate data from a known process (e.g., true ATT = 5, true coefficient = 1) and verify that the estimator recovers the parameter within tolerance, that pre-trends are null when

Table 8: Test distribution by module category

Module	Tests	Validation ^a	Methods
Regression (OLS, diagnostics, NLS, LOESS, GLS)	120	55	25
Panel data (FE, RE, HDFE, GMM)	95	40	12
Causal inference (DiD, IV, RD, synth)	130	52	20
Treatment effects (IPW, TMLE, DML, matching)	145	60	18
Discrete choice (logit, probit, count)	55	22	9
Time series (VAR, ARIMA, GARCH, Kalman)	110	45	20
Statistical tests	160	65	50+
ML (clustering, PCA, t-SNE, SVM)	120	35	10
Spatial econometrics	50	18	8
Survival analysis	35	12	5
Data, simulation, export, visualization	148	38	—
Total	2,168	442	250+

^a Tests that compare against R reference implementations or verify recovery of known parameters from simulated data-generating processes.

they should be, that weights sum correctly, and that confidence intervals have appropriate coverage. The corresponding R implementations (e.g., **did**, **DoubleML**, **Synth**) use different default tuning parameters, cross-validation splits, and optimization algorithms that complicate exact numerical comparison. Extending exact cross-validation coverage to these methods is a priority for future releases.

Appendix E documents tolerance thresholds by method category and sample size. Appendix F lists known differences from reference implementations (default robust SE type, convergence tolerances, R^2 variant selection)—none affecting the validity of statistical inference.

For core panel methods, **prompt2analytics** provides comparable coverage to specialized R packages; the primary remaining gap is Conley spatial standard errors. Users requiring maximum performance for high-dimensional fixed effects should note that **fixest** (Bergé 2018) likely outperforms **prompt2analytics** due to its highly optimized C++ implementation. The distinguishing features are the natural language interface and fully local execution without R dependencies; the package complements rather than replaces specialized tools. Having established the system’s accuracy and feature coverage, the next section examines whether it can run efficiently—and entirely locally—on consumer hardware.

6. Performance and deployment

The three principles require software fast enough to run locally and models small enough to fit on consumer hardware. This section presents performance benchmarks against R and describes fully local deployment configurations where all components—including the language model—run on the user’s machine.

6.1. Performance comparison with R

Table 9 presents benchmark results from a unified validation and benchmarking pipeline that simultaneously measures correctness, execution time, and memory usage. The pipeline covers 384 benchmark configurations across 140+ unique methods, with sample sizes ranging

from $n = 50$ to $n = 10,000$ (see Appendix E for details). Both R and Rust operate on identical datasets generated with shared random seeds; timings measure computation only, excluding data loading and process startup. R benchmarks use canonical reference packages: `lm()`/`sandwich` (Zeileis 2004) for regression, `plm` (Croissant and Millo 2008)/`lfe` (Gaure 2013) for panel data, `AER` for IV, `forecast` (Hyndman and Khandakar 2008)/`vars` for time series, and R’s `stats` for hypothesis tests. Across all 384 configurations (multiple sample sizes per method), the median speedup is $25.1\times$, with Rust faster in 361 cases (94%). At the largest sample size per unique method (119 methods), the median is $11.0\times$ (Figure 6).

Speedups vary widely by method category, reflecting the relative weight of interpreter overhead versus numerical computation. *Causal inference* methods show the largest gains: IV/2SLS ($146\times$) and difference-in-differences ($81\times$). *Treatment effect* estimators are similarly fast: TMLE ($283\times$), CBPS ($198\times$), and IPW ($171\times$). *Panel data* methods benefit from avoiding formula parsing and factor handling: fixed effects ($207\times$), HDFE ($176\times$), and random effects ($136\times$). *Time series* methods show moderate to large gains: VECM ($56\times$), ARIMA ($37\times$), and Granger causality ($4\times$). *Statistical tests* cluster around $20\text{--}40\times$: t -tests ($40\times$), ANOVA ($31\times$), and Shapiro-Wilk ($30\times$). *Regression* methods range from $13\times$ (OLS) to $27\times$ (quantile regression), with robust standard errors at $18\times$ (HC1). *Discrete choice* models range from $8\times$ (logit) to $15\times$ (Poisson). *Survival* methods achieve $14\text{--}19\times$ (Cox PH $14\times$, Kaplan-Meier $19\times$). *Machine learning* methods show more modest gains of $2\text{--}3\times$ (K-means $3\times$, PCA $3\times$, DBSCAN $2\times$), reflecting the competitiveness of R’s underlying C implementations for these algorithms.

Complete benchmark results are in Appendix E; reproduction instructions are in Appendix D.

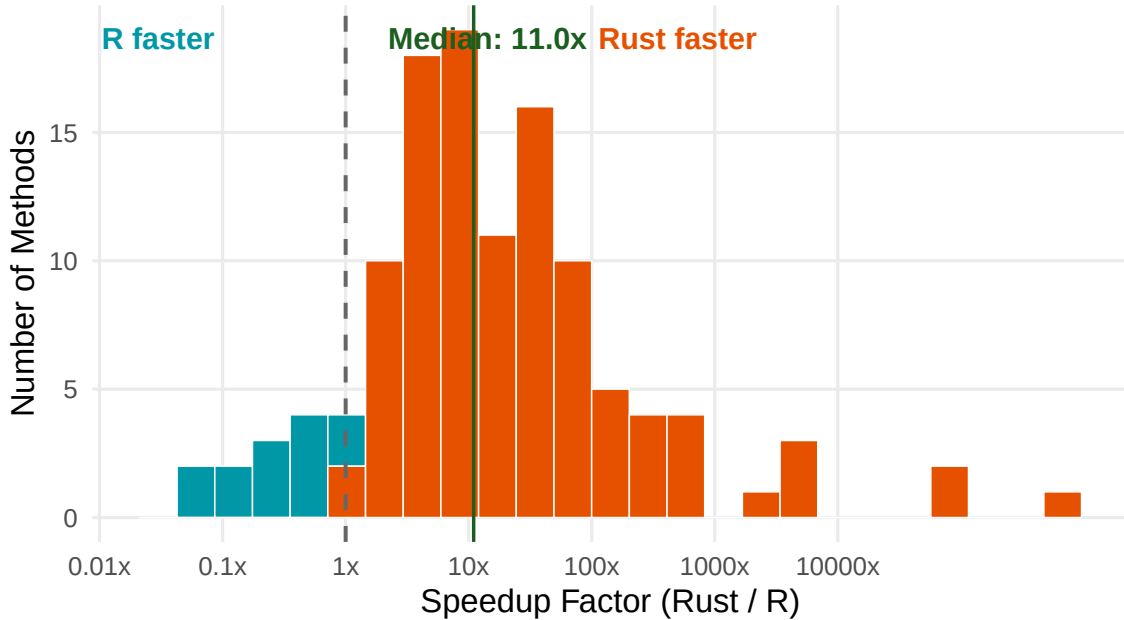


Figure 6: Distribution of Rust/R speedup factors across 119 unique methods at their largest sample size (log scale). The vertical line at $1\times$ indicates parity; values to the right indicate Rust faster. The median speedup is $11.0\times$, with Rust faster in 89% of methods.

Table 9: Performance benchmark results across representative methods (median execution time). Speedup = R time / Rust time. Sample sizes vary by method to reflect typical use cases ($n = 100$ – $10,000$). Memory columns show per-call allocation from R and Rust respectively.

Category	Method	n	R (μ s)	Rust (μ s)	Speedup	R Mem	Rust Mem
Causal	IV/2SLS	1,000	3,545	60	$58.8\times$	496.5 KB	184.8 KB
Causal	DiD	1,000	2,407	87	$27.6\times$	283.3 KB	128.6 KB
Discrete	Logit	1,000	5,269	462	$11.4\times$	1.8 MB	448.9 KB
Discrete	Probit	1,000	3,028	745	$4.1\times$	1.6 MB	417.5 KB
Discrete	Poisson	1,000	1,899	878	$2.2\times$	2.4 MB	2.4 MB
ML	K-Means ($k = 3$)	5,000	25,037	2,200	$11.4\times$	3.2 MB	589.0 KB
ML	PCA	5,000	1,820	251	$7.3\times$	1.3 MB	232.6 KB
ML	DBSCAN	5,000	305,585	104,956	$2.9\times$	22.1 KB	314.6 MB
Panel	FixedEffects	5,000	22,260	722	$30.8\times$	4.8 MB	858.8 KB
Panel	Hausman	5,000	97,004	3,510	$27.6\times$	22.1 MB	2.7 MB
Panel	RandomEffects	5,000	51,148	2,516	$20.3\times$	17.3 MB	1.8 MB
Panel	HDFE (2-way)	5,000	16,823	2,355	$7.1\times$	3.1 MB	1.1 MB
Regression	OLS + HC3	10,000	47,691	1,889	$25.2\times$	10.4 MB	1.3 MB
Regression	OLS	10,000	37,579	1,753	$21.4\times$	3.6 MB	1.1 MB
Regression	LOESS	1,000	9,071	1,749	$5.2\times$	293.0 KB	91.8 KB
Regression	GLS	1,000	311,119	812,183	$0.4\times$	805.3 MB	30.8 MB
Regression	NLS	1,000	3,367	38,448	$0.1\times$	—	—
Stats	t -test	1,000	45	5	$9.3\times$	23.7 KB	17 B
Stats	ANOVA	1,000	980	135	$7.3\times$	240.3 KB	55.6 KB
Survival	Cox PH	1,000	6,705	354	$18.9\times$	1.2 MB	349.7 KB
Survival	Kaplan–Meier	1,000	2,432	222	$11.0\times$	839.0 KB	391.9 KB
Time Series	ARIMA(1,1,1)	500	8,945	291	$30.8\times$	607.5 KB	591.4 KB
Time Series	MSTL	500	4,305	157	$27.4\times$	409.0 KB	47.5 KB
Time Series	Holt–Winters	500	9,621	1,078	$8.9\times$	7.4 MB	4.4 MB
Treatment	IPW	1,000	1,260,952	311	$>1000\times$	422.6 MB	325.2 KB
Treatment	Matching	1,000	15,204	1,616	$9.4\times$	2.2 MB	647.5 KB

The performance advantages stem from Rust’s ownership system (stack allocation and in-place mutation without garbage collection), the **faer** library’s SIMD-optimized linear algebra, the column-based API that eliminates formula parsing, and the single-language design that avoids data serialization between components. **prompt2analytics** performs best on computationally intensive iterative methods (panel fixed effects, treatment effect estimators, causal inference, robust standard errors) where compile-time optimizations and memory efficiency compound across iterations. For machine learning methods dominated by optimized C code (PCA, K-means, DBSCAN), R’s compiled implementations provide competitive performance.

Memory efficiency. Beyond execution time, Rust’s deterministic memory management reduces heap allocation. Across the 384 benchmark configurations, Rust typically allocates 2–50× less memory than the equivalent R call. The largest savings appear in panel data and treatment effect methods, where R’s formula interface and factor expansion create intermediate copies that Rust’s ownership model avoids. Figure 7 and Table 10 show the memory comparison by module.

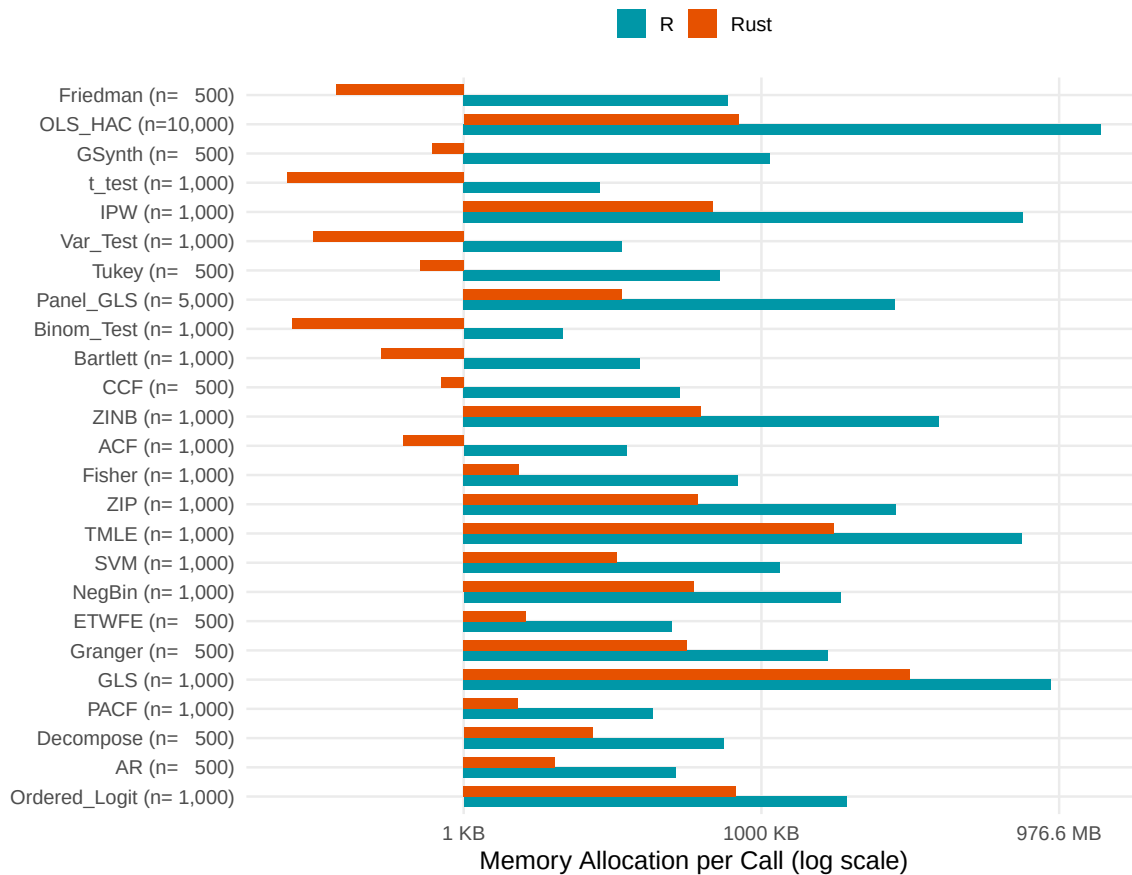


Figure 7: Memory allocation comparison between R and Rust implementations by method category. Rust consistently allocates less memory, with the largest savings in panel data and treatment effect methods.

For users with CUDA-capable hardware, **p2a-core** provides optional GPU acceleration via

Table 10: Memory allocation comparison by module (at largest sample size per method). Shows median per-call allocation for R and Rust, and the ratio (R / Rust).

Module	Methods	Median R Alloc	Median Rust Alloc	Median Ratio
Causal	4	389.9 KB	66.4 KB	16.0×
Discrete	9	6.1 MB	448.9 KB	7.8×
Treatment	6	28.4 MB	1.3 MB	6.9×
Regression	13	3.6 MB	1.2 MB	4.8×
Panel	8	11.1 MB	995.6 KB	4.6×
Time Series	11	409.0 KB	76.3 KB	4.1×
Stats	49	143.9 KB	39.1 KB	3.3×
Survival	4	1.0 MB	370.8 KB	3.0×
ML	9	3.2 MB	3.9 MB	1.5×
Overall	113	578.2 KB	204.1 KB	3.5×

cuBLAS and cuSOLVER. The `cuda` feature flag enables GPU code paths at compile time; at runtime, a singleton context attempts CUDA initialization and falls back silently to CPU if no device is available. Public function signatures are unchanged—users do not modify their code.

Not all operations benefit from GPU offloading. Transfer overhead between host and device memory dominates for small problems, and bandwidth-bound operations (e.g., matrix-vector products) can be faster on CPU even for large n . The dispatch logic therefore checks two conditions before routing to GPU: a minimum problem dimension (e.g., $k \geq 30$ columns for $\mathbf{X}'\mathbf{X}$) and a minimum computational intensity (e.g., $nk^2 \geq 2 \times 10^7$). Both thresholds are configurable via environment variables, allowing users to tune dispatch for their specific hardware. A shape guard additionally prevents GPU dispatch for tall-skinny matrix multiplications, where the CPU path is substantially faster.

Table 11 summarizes the operations with GPU code paths and observed speedups on an NVIDIA DGX Spark (Grace Blackwell, 12 GB VRAM, unified memory). The largest gains appear in $\mathbf{X}'\mathbf{X}$ computation (up to 7×

Table 11: GPU-accelerated operations and observed speedups

Operation	Use in	Speedup	Condition
$\mathbf{X}'\mathbf{X}$ (DGEMM)	OLS, panel FE/RE, IV, robust SEs	2–7×	$k \geq 30$
PCA covariance	Dimensionality reduction	4–13×	$n > 4k$
K-means distances	Clustering	2–3×	$d \geq 20$
Sandwich $\mathbf{X}'\mathbf{\Omega}\mathbf{X}$	HC0–HC3 robust SEs	2–3×	$k \geq 30$

Note: Measured on NVIDIA DGX Spark (Grace Blackwell, unified memory). Speedups are relative to the CPU path on the same machine (ARM Cortex-X925 with OpenBLAS). Results will differ on discrete GPUs where host–device transfer costs are higher.

Most econometric applications involve moderate numbers of regressors ($k < 30$), where CPU execution is faster. GPU acceleration is therefore not part of the default build; it targets high-performance computing scenarios with many covariates, high-dimensional clustering, or large-scale bootstrap procedures. The implementation exploits a layout convention—`ndarray`’s row-major storage appears as a transposed column-major matrix to cuBLAS—that avoids data reordering on the host side. Operations that are bandwidth-bound rather than compute-bound ($\mathbf{X}'\mathbf{y}$ via DGEMV) remain on CPU, where OpenBLAS is consistently faster. The feature is currently limited to NVIDIA GPUs via CUDA; extension to other backends (e.g., Vulkan compute, Metal) is left for future work.

6.2. Fully local deployment

While cloud-hosted language models offer convenience, many use cases require complete data isolation. The fully local configuration runs all components on the user’s machine via Ollama (Ollama Inc. 2024), eliminating all external network communication. Appendix B provides platform-specific installation instructions.

The evaluation results (Section 5) inform practical model recommendations. Open-source models deployable via Ollama—such as Ministral 3B (80.2% adequate) and Llama 4 Scout (50.0%)—enable fully local analysis without cloud connectivity, and Ministral 3B’s 80% adequate rate shows that data isolation need not come at the cost of tool selection quality for standard workflows.

Table 12 summarizes model options with hardware requirements and recommended use cases. Cloud-hosted models (GPT-4.1 Mini, Sonnet 4.6, Haiku 4.5, Gemini 2.5 Flash) provide higher accuracy but require API access; the table focuses on locally deployable options.

Table 12: Local deployment: models, hardware, and use cases

Model	Params	Storage	RAM	Adeq. (%)	Recommended for
<i>Evaluated models (locally deployable via Ollama)</i>					
Llama 4 Scout	17B ^a	12 GB	16 GB	50.0	Simple single-turn analyses
Ministral 3B	3B	2.0 GB	6 GB	80.2	General local use
<i>Additional local options (not in our evaluation)</i>					
Llama 3.3 70B	70B	40 GB	48 GB	— ^b	Multi-turn workflows
Qwen 2.5 7B	7B	4.4 GB	8 GB	— ^b	Balanced accuracy

Note: Adequate rate from the 96-prompt end-to-end evaluation (Section 5). Storage refers to Q4_K_M quantization (4-bit). RAM is memory during inference, typically 1.5–2× storage due to KV cache and computational buffers. ^a 17B active parameters, 109B total (mixture of experts). ^b Not included in our evaluation; recommended based on external tool-calling evaluations.

Ministral 3B demonstrates that even very small models achieve viable accuracy. With only 3 billion parameters, it runs on virtually any modern laptop—approximately 6 GB VRAM for GPU inference, or CPU-only with 8 GB RAM. Its adequate rate (80.2%) is within 4 percentage points of Claude Haiku 4.5 (81.2%), a cloud-hosted model. The 7–8B parameter class offers a practical middle ground, fitting on consumer GPUs with 8+ GB VRAM (e.g., NVIDIA RTX 3070/4070) or Apple M-series devices with 16 GB unified memory.

Beyond LLM memory, users must budget RAM for dataset operations. A typical configuration analyzing 100,000 observations with a 7B model requires approximately 8–10 GB total RAM; a minimal laptop setup (Minstral 3B, small datasets) works within 16 GB. Appendix B.2 provides detailed RAM requirements by dataset size (Table B.1) and system configurations for common deployment scenarios (Table B.2).

6.3. Privacy guarantees

The fully local configuration provides complete data isolation. No network requests are made to external services. All model inference occurs on the user’s hardware, dataset contents never leave the machine, and query history remains in local storage. This architecture is suitable for analysis of sensitive data where regulatory requirements (GDPR, HIPAA, or institutional review board protocols) or organizational policy prohibit cloud processing. Users should verify that their Ollama installation does not enable telemetry or automatic model updates. Appendix C provides detailed data transmission documentation and security best practices.

7. Discussion

This paper described chat-first data analytics—separating interpretation from computation, pre-validating methods rather than generating code, and keeping data local—and implemented the approach in **prompt2analytics**, a Rust library of over 200 validated methods exposed through MCP. The evaluation (Section 5) shows that tool selection works well: 81–91% adequate rate for frontier models, with errors concentrated in edge cases requiring fine distinctions between related methods.

Several limitations qualify these results. The practical benefits of chat-first interfaces—accessibility, reduced cognitive load, exploratory dialogue—remain hypothesized, as no user studies were conducted. Controlled experiments comparing task completion time, error rates, and user satisfaction against traditional code-based workflows are needed and represent the most important direction for future research. When the correct tool is selected, manual inspection of recorded tool calls shows that core parameters are almost always correct; the main imprecision is omission of optional arguments that fall back on sensible defaults. Users should verify that the system ran the analysis they intended, not merely that it selected the right method. While method coverage is broad, important categories remain unimplemented—Bayesian inference, Heckman selection, Conley spatial standard errors, and structural equation modeling among them (Table A.1 in Appendix A documents gaps and recommends alternatives). Data are loaded entirely into memory, limiting applicability to datasets that fit in RAM. And because the MCP schema uses semantic versioning (currently 0.x), the API may change as the software matures. We follow a conservative deprecation policy: tool names and required parameters will not be removed or renamed without a major version increment; new optional parameters can be added in minor releases. Tool call transcripts from the examples in this paper will remain valid in all 0.x releases. The underlying MCP protocol (**rmcp** 0.8) is itself pre-1.0; protocol-level changes are absorbed by the server without affecting tool schemas.

These limitations suggest clear directions for future work. Reducing omission of optional parameters—through structured output constraints, two-pass verification where the model checks its own extraction before execution, or richer schema descriptions that make optional parameters more salient—would narrow the gap between correct tool selection and fully spec-

ified analysis execution. Property-based testing could strengthen correctness guarantees beyond comparison to reference implementations; QuickCheck-style testing for statistical properties (e.g., residuals orthogonal to regressors) would complement numerical validation.

Foreign function interfaces to R and Python would let users call the Rust implementations from familiar environments, combining compiled performance with existing analytical workflows. An alternative design would have been to build **p2a-core** as an R package from the start, leveraging R’s ecosystem and user base. We chose a standalone architecture because the primary interface is the MCP server—a long-running process that any LLM client can connect to—which does not fit naturally into R’s package model. FFI wrappers remain a priority for giving R and Python users direct access to the compiled implementations.

If extraction quality improves sufficiently, the bottleneck shifts from “did the model understand the request?” to “does the method library cover this analysis?”—at which point extending coverage becomes more valuable than improving the language model, which is exactly the division of labor the architecture is designed to exploit.

Even with these extensions, the system’s role would remain complementary. **prompt2analytics** is not a replacement for R, Stata, or Python. It occupies a different niche: users who know what analysis they want but not which software command produces it. A researcher fluent in R has no need for a conversational interface to OLS; a policy analyst who has never used R might. The system complements existing tools rather than competing with them—and for users who outgrow the conversational interface, every tool call maps to a public function that can be called directly from Rust or, once foreign function interfaces are available, from R and Python.

Maintaining a library of over 200 validated methods is a legitimate concern. The same AI-assisted development tools that motivate the project also address this: coding agents can implement new methods, write validation tests, update dependencies, and respond to bug reports filed as GitHub issues—with human review of each pull request ensuring correctness. This workflow already produced the majority of the current implementation. It scales sublinearly with method count because new methods reuse established patterns (the `LinearEstimator` trait, shared matrix operations, the MCP handler template), and because validation against R reference implementations is itself automatable. The project is open-source under the MIT license; community contributions follow the same review process.

More broadly, the chat-first approach exemplifies a pattern applicable beyond econometrics: using language models as interpreters between human intent and specialized software (Yao *et al.* 2023; Schick *et al.* 2023). Any domain where users express goals in natural language and systems execute well-defined operations—laboratory automation, geographic information systems, circuit simulation—could adopt the same architecture, particularly as structured protocols like MCP (Anthropic 2024) standardize the interface between models and tools.

The LLM landscape evolves rapidly. The 96-prompt end-to-end evaluation (Section 5.3), which scores the full pipeline including tool selection, parameter extraction, numerical correctness, and interpretation, shows adequate rates of 50–91% across six models in single-turn mode. However, the Chrome multi-turn evaluation demonstrates that the deployed system—with dataset context, conversation history, and iterative refinement—achieves 96.7% accuracy (232/240 points) with Claude Sonnet 4.6, suggesting that the gap between isolated and deployed performance is substantial and that infrastructure quality matters as much as model capability. The evaluation harness and test cases are open-source, so new models can be

benchmarked as they appear. Importantly, improving LLM capability does not undermine the architecture—it strengthens it. Better models reduce parameter omissions and may eventually make code-generation approaches more reliable, but the benefits of pre-validation and data locality are orthogonal to model quality. Even if future models generate flawless code, users who require auditable, reproducible analyses on local data would still benefit from fixed, validated tool implementations.

The viability of local LLM deployment shown in Section 6 has implications beyond data privacy. If small models can reliably select among pre-implemented methods, conversational interfaces to scientific software become possible without cloud connectivity or high-end hardware—extending the separation principle (Section 2) to settings where code-generation approaches would require both model capability and network access that may not be available.

Acknowledgments

The author thanks the reviewers for constructive feedback that improved the paper.

Appendix

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- B Deployment guide
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A. Best practices and coverage gaps

This appendix provides practical guidance for effective use of **prompt2analytics** and documents method coverage gaps.

A.1. Pre-analysis checklist

Before running an analysis through **prompt2analytics**, users should verify:

1. *Method availability*: Confirm the required method is implemented. Table A.1 documents coverage gaps; Appendix F provides method-specific caveats.
2. *Data requirements*: Verify the dataset meets method assumptions. Panel methods require entity and time identifiers; instrumental variables (IV) require instruments; difference-in-differences (DiD) requires treatment and time indicators.
3. *Variable naming*: Use clear, unambiguous column names. The LLM extracts variable names from natural language; names like `x1` or `var_1` are more error-prone than `wage` or `education`.
4. *Specification clarity*: Specify methods explicitly when precision matters. “Estimate a within-transformation fixed effects model with clustered standard errors at the firm level” yields more reliable results than “run a regression with firm effects.”

A.2. Recommended workflows

The system is most effective for:

- *Exploratory analysis*: Quickly testing specifications before committing to a formal analysis pipeline.
- *Teaching and learning*: Students can explore econometric methods conversationally while seeing which tools and parameters are invoked.

- *Rapid prototyping*: Researchers can iterate on model specifications in natural language before writing production code.
- *Verification*: Cross-checking results from other software by running the same analysis through an independent implementation.

Users should prefer R, Stata, or Python when methods are unavailable (Bayesian inference, Heckman selection), maximum performance is critical (**fixest** outperforms for high-dimensional fixed effects (HDFE) at scale), reproducibility requires code scripts, or publication-quality visualizations are needed.

A.3. Method coverage gaps

While method coverage is broad (over 200 methods), several important categories remain unimplemented. Table A.1 summarizes coverage gaps and recommends alternative tools.

Table A.1: Method coverage gaps

Method Category	Status	Recommended	Alternative
Bayesian inference	Not implemented	R: rstanarm , Python: PyMC	brms ;
Heckman selection	Not implemented	R: sampleSelection ;	Stata: heckman
Conley spatial SEs	Not implemented	R: fixest , lfe	
Full SARIMA (arbitrary seasonal)	Partial support	R: forecast	
Quantile IV	Not implemented	R: quantreg	
Structural equation modeling	Not implemented	R: lavaan	
Weak instrument inference	F -stat only; no Stock-Yogo CVs	R: ivmodel , AER	

Note: Lists unimplemented or partially implemented method categories with recommended alternatives in R, Stata, or Python.

Users encountering these gaps should consult the listed alternatives; future releases aim to address Bayesian inference and Heckman selection as priorities.

B. Deployment guide

Full platform-specific installation instructions (Linux, macOS, Windows/WSL2, Docker) are maintained in the repository's `README.md` at <https://github.com/umatter/prompt2analytics>. Pre-compiled binaries for all major platforms are available on the GitHub releases page. This appendix documents only the MCP client configuration and hardware requirements referenced from Section 6.

B.1. MCP client configuration

For Claude Desktop, add to `claude_desktop_config.json`:

```
{
  "mcpServers": {
    "prompt2analytics": {
      "command": "/path/to/p2a-mcp",
      "args": ["--transport", "stdio"]
    }
  }
}
```

Other MCP-compatible clients (Cursor, Windsurf, custom agents) use the same `-transport stdio` interface. For HTTP-based integration, start the server with `-transport http -port 8080`.

B.2. Hardware requirements for data processing

Beyond LLM memory, users must budget RAM for dataset operations. Rust processes data in compact columnar format via **polars**. The primary memory consumers are design matrices ($8nk$ bytes for n observations and k predictors), covariance matrices ($8k^2$ bytes), bootstrap samples ($8kB$ bytes for B replications), and panel transformations (approximately equal in size to the original dataset).

Table B.1: RAM requirements for data processing

Rows	File size	Base RAM	Matrix ops	Notes
1,000	<1 MB	50 MB	+20 MB	Standard laptop
10,000	1–10 MB	100 MB	+100 MB	Standard laptop
100,000	10–100 MB	500 MB	+500 MB	Consumer hardware
1,000,000	100 MB–1 GB	2 GB	+2 GB	Workstation
10,000,000	1–10 GB	10 GB	+8 GB	High-memory system

Note: Peak memory during analysis. Assumes typical econometric datasets with 10–50 columns.

For combined LLM and data processing, sum the requirements from Tables 12 and B.1. A typical configuration analyzing 100,000 observations with a 7B model requires approximately 8–10 GB total RAM. Table B.2 provides system recommendations for common deployment scenarios.

Table B.2: System configurations by deployment scenario

Scenario	Model	RAM	GPU VRAM	Storage
Minimal (laptop)	Minstral 3B	16 GB	6 GB	5 GB
Consumer (desktop)	Llama 4 Scout	16 GB	12 GB	15 GB
Professional	Llama 3.3 70B	64 GB	48 GB	50 GB
Server (large data)	Cloud API	32 GB	—	2 GB

Note: RAM includes both LLM and data processing headroom. GPU requirements assume CUDA-capable NVIDIA cards; AMD and Intel GPUs are supported via Ollama’s ROCm and OneAPI backends.

For the simplest setup, the Docker configuration (Appendix B) bundles all dependencies including a pre-downloaded model and requires no local toolchain installation.

C. Privacy and security

This appendix documents data transmission behavior and security considerations.

C.1. Data transmission summary

Table C.1 lists operations that may transmit data beyond the local machine when using cloud LLM providers.

Table C.1: Data transmission by operation type

Operation	Data Transmitted	Risk Level
Dataset load	Column names, types, row count, null percentages	Low
Data preview (head/tail/sample)	Actual row values	High
Data profiling	Frequent values, min/max, examples	Medium
Statistical estimation	Coefficients, SEs, test statistics only	Low
Outlier detection	Specific observation values	Medium
Error messages	May include problematic values	Low

Note: Applies only when using cloud LLM providers. Local deployment via Ollama eliminates all external data transmission.

C.2. Audit logging

To enable audit logging of all MCP tool calls:

```
p2a-mcp --transport stdio --log-level debug --log-file audit.log
```

The log records: timestamp, tool name, parameters (excluding bulk data), and result summary. Review logs to verify no unexpected data transmission.

C.3. Network verification

To verify local-only operation with Ollama:

1. Start Ollama: `ollama serve`
2. Block outbound connections: `sudo ufw default deny outgoing`
3. Run analysis—it should complete successfully
4. Restore network: `sudo ufw default allow outgoing`

C.4. Security best practices

- Avoid loading untrusted data files (potential injection payloads)
- Review tool calls when processing sensitive data

- Use local LLM deployment for high-security environments
- Keep software updated; security patches are released as needed
- The MCP server has no filesystem access beyond loaded datasets

The primary mitigation for data privacy concerns is local LLM deployment via Ollama, which eliminates all external data transmission while maintaining full analytical capability.

D. Reproducibility

This appendix provides detailed instructions for reproducing the results in this paper.

D.1. Version information

Results in this paper were generated with the following software versions:

- Rust: 1.92.0 (stable)
- R: 4.5.2 with OpenBLAS
- Key R packages: **plm** 2.6-4, **lfe** 3.1-0, **sandwich** 3.1-1, **fixest** 0.12.1, **forecast** 8.23.0
- Operating system: Ubuntu 24.04 LTS (Linux 6.8)

D.2. Reproducing benchmarks

Performance benchmarks can be reproduced using the following protocol:

1. Install R 4.5+ with required packages: **plm**, **lfe**, **sandwich**, **forecast**, **bench**
2. Build **prompt2analytics** in release mode: `cargo build -release -p p2a-cli`
3. Execute benchmarks: `make -C paper benchmark-full`

Both languages use `seed = 42` for random number generation. The benchmark protocol includes 10 warmup iterations before 100 measured iterations for fast methods (50 for moderate, 20 for slow). Detailed instructions: `paper/code/README_REPRODUCIBILITY.md`.

D.3. Containerized reproduction

The repository's `docker-compose.yml` provides a containerized deployment of the MCP server. For reproducing the paper's tables and figures, the native toolchain described above is recommended; a containerized paper build requires a multi-stage image with both R 4.5.2 (plus packages listed above) and Rust 1.92.0. A sample `Dockerfile` for this purpose is provided in `paper/Dockerfile.reproduce`; after building, run:

```
docker build -f paper/Dockerfile.reproduce -t p2a-paper .
docker run -v $(pwd)/paper:/output p2a-paper make all
```

Results should match within the tolerances specified in Appendix E.

E. Benchmark methodology

This appendix provides complete details on the benchmark methodology used in Section 6.

E.1. Data generating processes

Benchmark datasets are generated synthetically using identical data-generating processes in both languages with fixed random seeds (seed = 42) for exact reproducibility.

For regression benchmarks, data follow $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$ with true parameters $\beta_0 = 1.0$, $\beta_1 = 2.0$, $\beta_2 = -1.5$, and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ where $\sigma = 1.0$; covariates are drawn as $x_{1i} \sim \mathcal{N}(0, 1)$ and $x_{2i} \sim \mathcal{N}(0, 1)$ with correlation $\rho = 0.3$. For panel data, entity effects $\alpha_i \sim \mathcal{N}(0, 0.5^2)$ are added, with 100 entities observed over 10 periods. For discrete choice, the latent index $y_i^* = \beta'x_i + \varepsilon_i$ is transformed via the logit or probit link. For time series, data follow ARIMA(1, 1, 1) with $\phi = 0.7$, $\theta = 0.3$, and innovation variance $\sigma_\eta^2 = 1.0$.

Complete DGP specifications are provided in `paper/code/README_REPRODUCIBILITY.md`.

E.2. Benchmark execution protocol

Each benchmark configuration executes multiple iterations after a warmup phase:

- 100 iterations for fast methods (OLS, robust SEs, PCA)
- 50 iterations for moderate methods (panel FE, clustering)
- 20 iterations for slow methods (ARIMA, MSTL (multiple seasonal-trend decomposition using LOESS))

R benchmarks use the **bench** package, which accounts for garbage collection timing. Rust benchmarks use a custom harness (`harness = false` in `Cargo.toml`) with automatic warmup and outlier detection. We report full timing distributions (minimum, quartiles, maximum, mean, standard deviation) rather than point estimates alone.

E.3. Hardware and software environment

All benchmarks run on identical hardware: an Intel Core i7-1260P processor with 64 GB RAM running Linux, with R 4.5.2 linked against OpenBLAS and Rust 1.92.0 compiled in release mode with link-time optimization. The **faer** library (version 0.22) used by **p2a-core** does *not* call an external BLAS; it implements its own GEMM microkernels in pure Rust with explicit SIMD via the **pulp** crate, dispatching at runtime to AVX2+FMA on x86-64 (or NEON on ARM). Both R and Rust therefore use optimized linear algebra routines, but from different implementations (OpenBLAS vs. **faer**). This makes the comparison a realistic measure of what each ecosystem delivers out of the box, not a controlled BLAS-versus-BLAS experiment.

E.4. Validation tolerances

Numerical validation against R reference implementations uses sample-size-dependent tolerances. We use absolute tolerances for quantities with known scale (e.g., test statistics) and relative tolerances for unbounded quantities (e.g., coefficient estimates).

Table E.1: Validation tolerances by method and sample size

Category	Quantity	Small ($n < 100$)	Medium	Large ($n > 10,000$)
<i>Coefficients and standard errors</i>				
OLS coefficients	Relative	10^{-6}	10^{-8}	10^{-10}
Robust SEs (HC0/HC1) ^a	Relative	10^{-6}	10^{-8}	10^{-8}
Robust SEs (HC2/HC3) ^a	Relative	10^{-5}	10^{-7}	10^{-7}
Panel FE/RE	Relative	10^{-5}	10^{-6}	10^{-6}
MLE ^b (logit/probit)	Relative	10^{-4}	10^{-6}	10^{-6}
<i>Test statistics and fit measures</i>				
t -statistics, F -statistics	Absolute	10^{-4}	10^{-6}	10^{-6}
R^2 , within R^2	Absolute	10^{-4}	10^{-6}	10^{-6}
p -values	Absolute	10^{-4}	10^{-6}	10^{-6}
Log-likelihood	Absolute	10^{-2}	10^{-4}	10^{-4}
<i>Hypothesis tests</i>				
Test statistics	Absolute	10^{-4}	10^{-6}	10^{-6}
Critical values	Absolute	10^{-4}	10^{-4}	10^{-4}

Note: Absolute tolerances (ε_{abs}) apply to test statistics, p -values, and bounded quantities. Relative tolerances (ε_{rel}) apply to coefficient estimates and standard errors. ^a HC0–HC3: heteroskedasticity-consistent standard error estimators (types 0–3). ^b MLE: maximum likelihood estimation.

Looser tolerances for small samples reflect increased sensitivity to algorithmic differences when matrices are ill-conditioned. HC2/HC3 standard errors use looser tolerances because leverage calculation involves $h_{ii} = x_i'(X'X)^{-1}x_i$, which accumulates rounding errors differently depending on whether leverage is extracted from QR decomposition (as in **sandwich**) or computed via direct matrix multiplication (as in **prompt2analytics**). Both approaches are numerically valid; the tolerance reflects implementation variation rather than error.

MLE methods use looser small-sample tolerances because convergence criteria differ between implementations: R's **glm()** uses relative change in deviance, while **prompt2analytics** uses absolute change in log-likelihood. Final parameter estimates can differ at the tolerance boundary while both satisfying their respective convergence criteria.

Tolerances were determined empirically by running validation tests across sample sizes from $n = 10$ to $n = 100,000$ and identifying the tightest tolerance that passes consistently. Edge cases (singular matrices, near-separation in logit/probit, small cluster counts) are tested separately with method-specific tolerances documented in the test code. All 416 validation tests pass at the specified tolerances.

F. Implementation notes

This appendix documents implementation caveats and limitations for specific methods.

F.1. Instrumental variables

The two-stage least squares (2SLS) implementation reports first-stage F -statistics for instrument relevance. The traditional rule of thumb ($F > 10$) from [Staiger and Stock \(1997\)](#) has been superseded by more nuanced guidance. [Stock and Yogo \(2005\)](#) provide critical values depending on the number of instruments and acceptable bias levels; [Lee, McCrary, Moreira, and Porter \(2022\)](#) offer further refinements for inference under weak identification. The current implementation does not provide Stock-Yogo critical values or weak-identification-robust inference methods (Anderson-Rubin, conditional likelihood ratio). Users with potentially weak instruments should verify results using R’s **ivmodel** or **AER** packages.

F.2. Difference-in-differences

The difference-in-differences (DiD) implementation assumes the canonical two-group, two-period design. Recent econometric literature has documented severe biases when this estimator is applied to staggered treatment timing. [Goodman-Bacon \(2021\)](#) demonstrates that two-way fixed effects DiD produces a weighted average of all possible 2×2 comparisons, including problematic “forbidden” comparisons using already-treated units as controls. [de Chaisemartin and D’Haultfoeuille \(2020\)](#) and [Sun and Abraham \(2021\)](#) propose alternative estimators robust to treatment effect heterogeneity.

prompt2analytics implements staggered DiD via Callaway-Sant’Anna ([Callaway and Sant’Anna 2021](#)), Goodman-Bacon decomposition, and extended two-way fixed effects (TWFE) ([Wooldridge 2021](#)). Users with complex staggered designs may also consult **fixest**’s `sunab()` function or the **did** package in R for additional diagnostics.

F.3. Random effects

The random effects estimator uses the Swamy-Arora variance component estimator, which can produce negative variance estimates for σ_α^2 (the between-entity variance) in certain data configurations. When this occurs, the implementation truncates the estimate to zero and proceeds with pooled OLS, issuing a warning.

F.4. Discrete choice models

Logit and probit models report coefficients on the latent index scale. For interpretation, marginal effects are computed as *Average Marginal Effects* (AME): the partial derivative $\partial \Pr(y = 1) / \partial x_j$ is evaluated at each observation’s covariate values, then averaged across the sample. This differs from *Marginal Effects at Means* (MEM), which evaluates derivatives at the sample mean of all covariates. AME is generally preferred because it does not depend on potentially unrepresentative “average” covariate values ([Cameron and Trivedi 2005](#)).

F.5. Time series models

Autoregressive integrated moving average (ARIMA) estimation supports non-seasonal models: $\text{ARIMA}(p, d, q)$ with automatic or user-specified orders. Seasonal ARIMA (SARIMA) is partially implemented: seasonal differencing and seasonal AR/MA terms can be specified, but automatic seasonal order selection is not available.

Vector autoregression (VAR) and vector error correction model (VECM) implementations

support lag selection via AIC/BIC and Johansen cointegration testing, but do not implement structural VAR identification.

F.6. Regression diagnostics validation

Diagnostic tests are validated against **lmtest** (Hothorn, Zeileis, Farebrother, and Cummins 2024): Jarque-Bera normality test statistics match to $< 10^{-4}$, Breusch-Pagan heteroskedasticity tests to $< 10^{-4}$, Durbin-Watson autocorrelation statistics to $< 10^{-6}$, and variance inflation factors (VIF) to $< 10^{-6}$.

F.7. Known differences from reference implementations

A small number of intentional differences exist between **prompt2analytics** and reference implementations. First, **prompt2analytics** defaults to HC1 robust standard errors (consistent with Stata’s **robust** option), while R’s **sandwich** defaults to HC3. Second, R uses variable names in output; **prompt2analytics** uses numeric indices internally but displays names in formatted output. Third, for panel models we report “within R^2 ” by default, matching **plm**’s convention for fixed effects models. Fourth, iterative methods (HDFE, maximum likelihood estimation) use a default convergence tolerance of 10^{-8} , which may differ slightly from R package defaults. These differences are documented in the method documentation and do not affect the validity of statistical inference.

Coverage gaps and unimplemented methods are documented in Appendix A (Table A.1).

G. Extended examples

This appendix collects additional examples that complement the core demonstrations in Section 4.

G.1. Comparison with R

For reference, the equivalent R code for the fixed effects estimation shown in Section 4.1 is:

```
library(plm)
data("Grunfeld", package = "plm")
fe_model <- plm(inv ~ value + capital, data = Grunfeld,
               index = c("firm", "year"), model = "within")
summary(fe_model)
```

The **prompt2analytics** implementation produces numerically equivalent results (Section 5), while offering a conversational interface for users unfamiliar with R syntax.

G.2. Time series analysis

The basic time series workflow (stationarity testing and ARIMA fitting) is demonstrated in Section 4.4. Here we note a limitation: order selection is delegated to the LLM rather than algorithmic search, so it is not equivalent to R’s `auto.arima()` (Hyndman and Khandakar 2008).

G.3. Handling messy data

Real-world datasets often require cleaning before analysis. The system detects non-standard missing indicators and guides type conversion:

User: Load survey_responses.csv and describe the data.

Dataset: survey_responses (2,847 rows x 6 columns)

Note: 'income' and 'satisfaction' are typed as String, likely due to non-standard missing value indicators (".", "N/A").

User: Replace "." and "N/A" with proper missing values, then convert income and satisfaction to numeric.

Cleaned: 89 values replaced across 2 columns.

income: now Float64, 52 missing (1.8%)

satisfaction: now Float64, 37 missing (1.3%)

The next example demonstrates a common challenge: ambiguous column names that do not match natural language descriptions. The dataset uses coded variable names rather than descriptive labels.

User: Load hr_extract_q3.csv and describe the data.

Dataset: hr_extract_q3 (1,204 rows x 8 columns)

emp_id: Int64 (1204 unique)

dept_cd: String (12 unique: "ENG", "MKT", "FIN", ...)

var_03: Float64 (range: 28000-185000)

var_04: Float64 (range: 0.5-35.2)

var_05: Int64 (0 or 1)

q3_rating: Float64 (range: 1.0-5.0, 23 missing)

loc_cd: String (8 unique: "NYC", "SFO", "CHI", ...)

dte_hire: String (formats vary: "2015-03-12", "03/12/2015")

User: var_03 is salary and var_04 is years of experience.

Regress salary on experience and department with robust SEs.

Renamed: var_03 -> salary, var_04 -> experience

Note: dept_cd is categorical (12 levels); using treatment coding with reference category "ENG".

OLS Regression (HC1 robust standard errors)

Dependent variable: salary

Observations: 1,204

...

This example illustrates both a strength and a limitation. The system correctly maps user-provided variable descriptions to coded column names and handles the categorical department variable automatically. However, the user must know what the coded variables represent—the system cannot infer that `var_03` is salary from its range alone. Inconsistent date formats in `dte_hire` (ISO 8601 and US format in the same column) would require explicit cleaning before use as a time variable.

H. End-to-end evaluation protocol

This appendix provides full protocol details for the end-to-end evaluation reported in Section 5. The automated evaluation harness drives the live **prompt2analytics** HTTP backend, loading datasets, sending prompts at varying clarity levels, and scoring complete responses including tool calls and LLM interpretations. Complete prompts and scoring data are provided in the supplementary materials.

H.1. Models

Table H.1 lists the six models evaluated, spanning frontier cloud, efficient cloud, and open-source deployment scenarios.

Table H.1: Models evaluated in end-to-end testing

Model	Provider	Purpose	Parameters
GPT-4.1 Mini	OpenRouter (OpenAI)	Frontier cloud	undisclosed
Claude Sonnet 4.6	Anthropic API	Balanced cloud	undisclosed
Claude Haiku 4.5	Anthropic API	Fast cloud	undisclosed
Gemini 2.5 Flash	OpenRouter (Google)	Fast cloud	undisclosed
Minstral 3B	OpenRouter	Minimal open-source	3B
Llama 4 Scout	OpenRouter (Meta)	Open-source MoE	17B active (109B total)

Note: All models accessed via API at temperature = 0. Open-source models (Minstral 3B, Llama 4 Scout) can alternatively be served locally via Ollama. Tool definitions auto-generated from MCP router schemas (119 Tier-1 tools).

H.2. Synthetic datasets

Six datasets with known data-generating processes provide ground-truth parameters for scoring. Table H.2 summarises each dataset.

H.3. Test design

Table H.3 summarises the test structure. Each of 32 core tests is written at three *clarity levels*: precise (names the exact method and all parameters), moderate (describes the goal without naming the method), and vague (loose natural language). This yields 96 single-turn prompts.

H.4. Scoring rubric

Each prompt is scored on four dimensions (Table H.4). A prompt is classified as *adequate*

Table H.2: Synthetic evaluation datasets

Dataset	Type	n	Key DGP features
<code>eval_cross_section</code>	Cross-section	500	$\text{salary} = 25000 + 3000 \cdot \text{edu} + 1500 \cdot \text{exp} + 500 \cdot \text{edu} \times \text{exp} + \epsilon$
<code>eval_panel</code>	Panel ($N=50, T=10$)	500	Firm and year FEs; $\beta_{\text{value}} = 0.1, \beta_{\text{capital}} = 0.05$
<code>eval_timeseries</code>	Time series	200	AR(2) with $\phi_1=0.6, \phi_2=-0.2$; trend; seasonal
<code>eval_treatment</code>	Treatment	2000	ATT = 3.0; confounding via x_1, x_2 ; 2-period DiD structure
<code>eval_messy</code>	Messy data	300	Coded names; -99 sentinels; "N/A" strings; mixed date form
<code>eval_survey</code>	Survey	400	Ordinal, binary, count outcomes; logit/probit/Poisson DGPs

Note: All datasets generated with `set.seed(42)` in R. True parameter values are documented in `PROTOCOL.md`.

Table H.3: Test categories and counts

Category	Tests	Clarity levels	Total prompts
Regression	5	3	15
Panel data	4	3	12
Causal inference	5	3	15
Time series	4	3	12
Hypothesis testing	4	3	12
Discrete choice	3	3	9
Machine learning	3	3	9
Messy data	4	3	12
Total	32		96

Note: Full prompts for all 96 tests are in `test_cases.json`.

if it scores $\geq 62.5\%$ of the applicable maximum (5/8 for numeric tasks, 4/6 when numerical correctness is N/A).

Table H.4: End-to-end scoring rubric

Dimension (0–2 each)	Scoring criteria
Tool selection	2 = exact correct tool; 1 = acceptable alternative; 0 = wrong tool or no tool call
Parameter extraction	2 = all parameters correct; 1 = core parameters correct, optional missing; 0 = wrong core parameters
Numerical correctness	2 = within tolerance of true DGP values; 1 = correct sign and magnitude; 0 = wrong; N/A for non-numeric tasks
Interpretation quality	2 = accurate with context; 1 = superficial but correct; 0 = misleading

Note: Maximum 8 points for numeric tasks, 6 for non-numeric. Adequate threshold: $\geq 62.5\%$ of applicable maximum.

H.5. Results by category

Table H.5 shows the adequate rate by test category and model. All scores use auto-generated tool schemas from the MCP router and the v2 evaluation harness with corrected parameter name matching. Hypothesis testing achieves 100% across the top four models. ML tasks are uniformly strong due to simpler parameter schemas. Causal inference shows the widest

variation (60–93%), reflecting the difficulty of distinguishing between IPW, matching, and DiD designs.

Table H.5: Adequate rate (%) by category and model

Category	GPT-4.1 Mini	Sonnet 4.6	Haiku 4.5	Ministr. 3B	Gemini 2.5 Fl.	Llama 4 Sc.
Hypothesis testing	100	100	100	75	100	75
Time series	100	100	92	100	67	42
Causal inference	93	73	33	87	60	60
Panel data	92	75	100	75	75	58
ML	89	78	89	78	89	67
Discrete choice	78	78	89	78	56	22
Regression	87	80	80	87	73	33
Messy data	83	92	83	58	83	42

Note: Adequate = $\geq 62.5\%$ of maximum score per prompt. Each category has 9–15 prompts (3 clarity levels per test case). Tool definitions auto-generated from MCP router schemas (119 Tier-1 tools).

H.6. Chrome multi-turn evaluation details

The Chrome-based evaluation drives the deployed Dioxus web frontend through automated browser interaction, testing the full pipeline including SSE streaming, session state management, and multi-turn context. Eight conversations with Claude Sonnet 4.6 cover the following workflows:

1. **MT1:** OLS regression $\rightarrow$ HC1 robust SEs $\rightarrow$ residual diagnostics (3 turns)
2. **MT2:** Panel data description $\rightarrow$ FE $\rightarrow$ Hausman test $\rightarrow$ RE (4 turns)
3. **MT3:** Time series description $\rightarrow$ stationarity tests $\rightarrow$ ARIMA $\rightarrow$ forecast (4 turns)
4. **MT4:** Naive OLS $\rightarrow$ DiD $\rightarrow$ balance check $\rightarrow$ IPW $\rightarrow$ comparison (5 turns)
5. **MT5:** Data quality assessment $\rightarrow$ sentinel replacement $\rightarrow$ renaming $\rightarrow$ regression (4 turns)
6. **MT6:** Survey data description $\rightarrow$ logit $\rightarrow$ ordered logit $\rightarrow$ negative binomial (4 turns)
7. **MT7:** OLS $\rightarrow$ diagnostic tests $\rightarrow$ specification choice (3 turns)
8. **MT8:** Cross-section OLS $\rightarrow$ panel FE $\rightarrow R^2$ comparison (3 turns)

Figure H.1 shows per-conversation scores. Five of eight conversations achieve perfect scores. The three imperfect conversations lose points to: data type casting limitations (MT5), superficial interpretation of confounding (MT4), and ordered logit convergence with unscaled predictors (MT6).

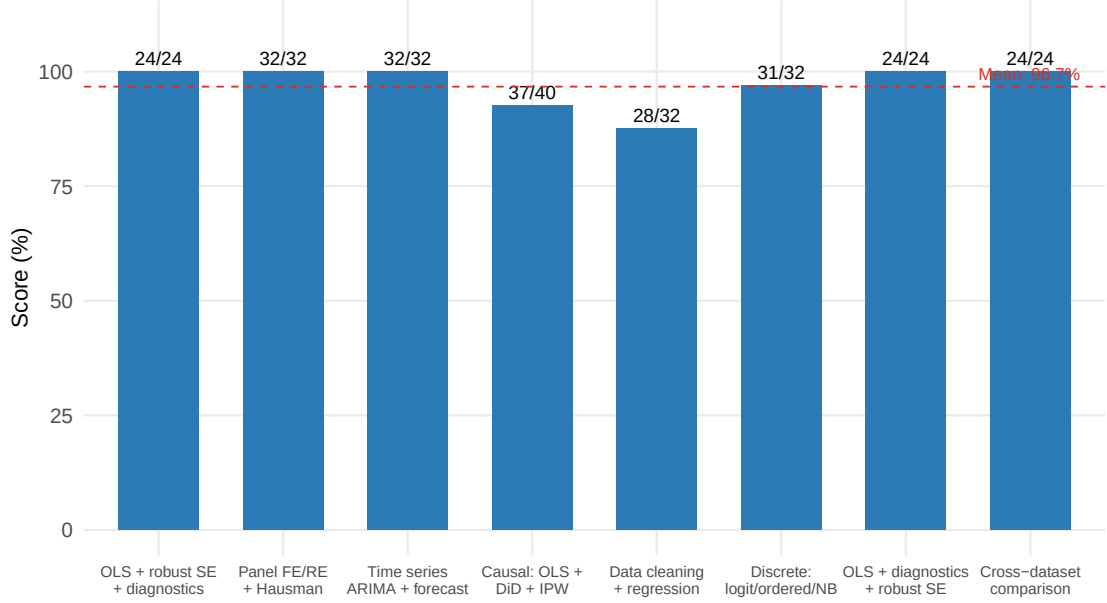


Figure H.1: Per-conversation scores in Chrome multi-turn evaluation (Claude Sonnet 4.6). Dashed line shows overall mean (96.7%).

Figure H.2 shows per-dimension averages. Tool selection and parameter extraction are perfect; numerical correctness (1.77/2.00) is the primary source of lost points, driven by tool limitations rather than model errors.

Table H.6: Infrastructure improvement trajectory

Stage	Fix description	Score	Accuracy
Pre-fix	API-only single-turn evaluation (baseline)	165/240	68.8%
Round 1	HTTP dispatch: replaced 70-tool match with macro-based router (268 tools)	220/240	91.7%
Round 2	LLM tool definitions: fixed formula $\rightarrow$ y/x for logit/probit/diagnostics/clustered/bgtest/driscoll_kraay; added ordered_model, negbin, str_replace_value, munge_rename, munge_mutate	232/240	96.7%

Note: Scores are for 8 multi-turn conversations (30 total turns, max 240 points) with Claude Sonnet 4.6 via Chrome. Round 1 replaced a manual 70-tool HTTP match with router dispatch (268 tools). Round 2 fixed LLM tool definitions and added missing tools.

H.7. Supplementary materials

The complete evaluation materials are in `paper/code/e2e_eval/`: `PROTOCOL.md` (full protocol), `test_cases.json` (all prompts with expected tools and parameters), `generate_datasets.R` (dataset DGPs), `run_evaluation.py` (automated evaluation harness), `generate_e2e_figures.R`

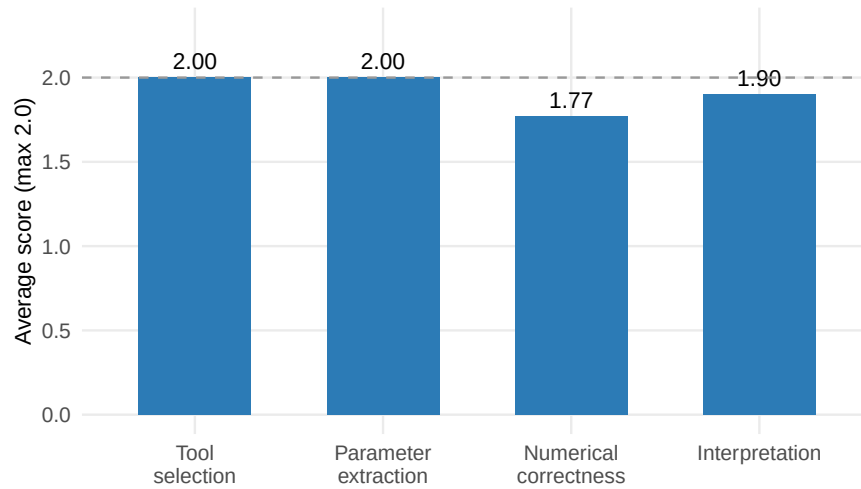


Figure H.2: Per-dimension scoring averages across 30 turns (Claude Sonnet 4.6, Chrome evaluation).

(figure and table generation from results), and **results/** (raw scoring data in JSON format for all six models, plus Chrome multi-turn results).

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